

BEYOND THE BASE: EXPONENT ALLOCATION IN ROPE

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ABSTRACT

Rotary position embeddings specify both a frequency range and an allocation of a finite number of rotary pairs within that range. We study the second choice separately from range selection and from the weights that learn to use the resulting basis. In paired 151.9M training runs with identical frequency endpoints, changing only the 30 interior exponents lowers extrapolation NLL at every tested extended length in all three seeds. The ordering reverses under a specified runtime range-retargeting policy, and weights-by-table crossings show strong dependence on the training allocation. We characterize the supplied positional basis through full sine-cosine subspaces, establish their shared slow-frequency limit, and derive a Cosh allocation from an explicit convex design criterion. Matched training and adaptation evaluate this construction beyond the controlled setting. For frozen deployment, we study native-relative exponent adjustments: a boundary-matched profile raises task-equal F1 from 21.62% to 25.44% on five OLMo long-context QA tasks, while a static Qwen table-amplitude configuration improves 64K RULER-13 by 6.09 points over the tested YaRN configuration. Together, these results identify interior allocation as a consequential design choice whose benefits depend on sampled range and learned compatibility.

1 INTRODUCTION

Rotary position embeddings (RoPE) assign a frequency to each two-dimensional query-key coordinate pair (Su et al., 2024). Its standard geometric table is parameterized by a base, controlling sampled positional scales (Liu et al., 2024; Xu et al., 2024). A finite table, however, makes another choice: how its rotary pairs occupy that range (Oka et al., 2026b; Tian et al., 2026). What remains necessary is to distinguish the causal effect of interior allocation from range selection, from the learning process, and from the compatibility of an installed table with mature weights.

Three questions organize the study. What changes in the positional basis when interior exponents are redistributed? Does this redistribution affect learned behavior when the frequency range is held strictly fixed? And how does an existing checkpoint respond to adjustments of its native exponents? These questions connect geometric analysis, controlled interventions, and practical table design.

Figure 1 shows the basic intervention: decompose the log-frequency table into endpoints (a , R) and normalized interior exponents z . In paired 151.9M training runs ($L_{\text{train}} = 256$, $K = 32$), holding endpoints identical while changing only the 30 interior exponents lowers extrapolation NLL across all three seeds. However, retargeting the runtime range reverses this ordering, and frozen weights strongly prefer runtime tables derived from their training allocation. Together, these controls isolate interior allocation as an independent design variable while establishing its action boundaries with range and learned representations.

The positional geometry provides a way to analyze these distributions. Each rotary pair supplies an entire sine-cosine subspace. Canonical correlations compare these subspaces without privileging a content phase, exposing the common low-frequency limit $\text{span}\{1, \Delta\}$. Thus many slow pairs supply few distinct positional directions over a bounded interval. The learned content coefficients determine how the model uses those directions.

We evaluate concrete allocation designs in two regimes. Before or during learning, a convex design criterion produces an analytic Cosh family whose quantiles give a fixed table, evaluated in scarce rotary channels, continued training, and matched 8B adaptation. For frozen checkpoints, we express modifications as displacements from native exponents. A static Qwen configuration improves 64K

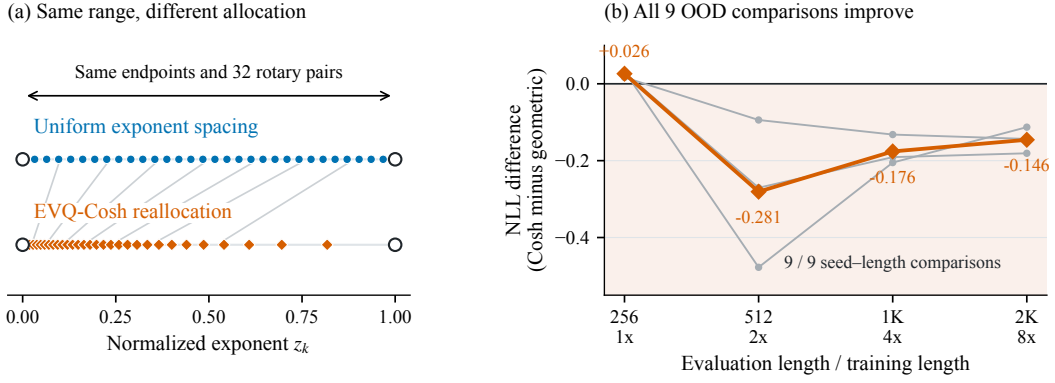


Figure 1: **Interior exponents change learned behavior at fixed frequency range.** Left: two allocations of the same 32 rotary pairs with identical endpoints. Right: paired 151.9M models trained at length 256; changing the 30 interior exponents lowers extrapolation NLL in every seed. Thin lines show the three training seeds and the heavy line their mean.

RULER-13 by 6.09 points over YaRN, and a boundary-matched adjustment raises OLMo mean token F1 from 21.62% to 25.44% across five natural-QA tasks (631 untruncated prompts). Checkpoint preferences differ across lengths, making learned compatibility central to allocation design.

Contributions. We (i) isolate interior exponent allocation from sampled range using paired training and frozen-model interventions, showing behavioral effects that cannot be explained by scalar range changes alone; (ii) characterize full sine-cosine subspace geometry, establish their shared slow-frequency limit, and delimit static metrics via non-identifiability and compatibility boundaries; and (iii) evaluate explicit allocation designs (analytic Cosh and boundary-matched) under matched learning and static deployment with complete-generation and natural-QA endpoints.

2 EXPONENT ALLOCATION

Let $K = d_{\text{rot}}/2 \geq 2$ be the number of rotary pairs. Standard RoPE uses

$$\omega_k = b^{-\phi_k}, \quad \phi_k = \frac{k}{K}, \quad k = 0, \dots, K-1, \quad (1)$$

for a base $b > 1$. For an arbitrary strictly ordered positive table $\omega_0 > \dots > \omega_{K-1} > 0$, write

$$x_k = -\log \omega_k = a + Rz_k, \quad z_k = \frac{x_k - x_0}{x_{K-1} - x_0}, \quad z_0 = 0, \quad z_{K-1} = 1. \quad (2)$$

The pair (a, R) specifies the sampled log-frequency support, while z specifies how the finite rotary budget occupies that support. Every geometric table has $z_k = k/(K-1)$, irrespective of its base. Thus changing z while fixing (a, R, K) is an intervention on interior allocation. The standard grid k/K in (1) and the endpoint-normalized shape $k/(K-1)$ are two descriptions of the same geometric table, not interchangeable installation rules.

A continuous density over normalized exponents can be discretized through its quantiles. At finite K , midpoint quantiles, endpoint-inclusive quantiles, and affine anchoring produce different installed frequency tensors. We specify the grid and anchoring convention for each protocol. In the fixed-support experiments, the nonuniform quantiles are affinely mapped to the baseline’s actual sampled endpoints. In mature checkpoints, we instead express an installed table through its displacement from the native tensor; §6 develops that form. These conventions prevent nominally identical bases or extension factors from concealing changes in the actual sampled range.

3 CONTROLLED EFFECTS OF EXPONENT ALLOCATION

We begin with interventions that distinguish exponent shape, sampled range, and the weights learned under an allocation. The Cosh quantiles used here are constructed in §4.4; endpoint anchoring keeps their range fixed. An exponential deformation supplies a second nonuniform family.

3.1 CONTROLLED EFFECT AT FIXED RANGE AND RANGE INTERACTION

We first ask whether interior allocation affects learned behavior when the sampled range is exactly fixed. We train paired 151.9M models with $K = 32$ rotary pairs at sequence length 256. Within each of three seeds, architecture, initialization, token order, optimizer, schedule, and the 499,974,144-token budget are identical. The geometric arm uses the FMRoPE training-base rule $b = L_{\text{train}}$ (Oka et al., 2026b); the Cosh arm anchors its $\tau = 4$ quantiles to the same endpoints. Only the 30 interior exponents differ.

At the fixed training support, the mean Cosh-minus-geometric tail-NLL differences at lengths 256, 512, 1024, and 2048 are $+0.026, -0.281, -0.176, -0.146$. All three seeds improve at each extended length (Fig. 1). Interior allocation thus changes learned generalization without any scalar base shift. However, retargeting the log-frequency span to each evaluation length reverses the ordering to $+0.060, +0.227, +0.460$ across all three seeds. This reversal demonstrates that allocation cannot be judged independently of operating range: favorable fixed-support orderings do not survive arbitrary range dilation (Appendix C).

3.2 CROSS-SHAPE FACTORIAL ACROSS CONFIGURATIONS

A 50.9M factorial varies two bases, two training lengths, three head dimensions, and three paired seeds, retaining endpoints throughout. At the reference Cosh strength, the weighted OOD objective improves in 7/12 configurations; preassigned $1.25\times$ improves 10/12, and matched exponential improves 9/12. The aggregate reference-Cosh-minus-exponential contrast is small and uncertain ($+0.00074$ NLL, 95% interval $[-0.006, 0.008]$; Table 16). These results establish allocation effects across configurations while bounding claims about the unique superiority of any single shape.

3.3 WEIGHT COMPATIBILITY AND TABLE CROSSINGS

The frequency table defines a basis in which query and key projections are learned. In a 50M pair with perplexities near 7, installing Cosh frequencies into Geo-trained weights raises PPL to 76.20; the reverse swap raises it to 23.05 (Fig. 2). A separate crossing at 151.9M repeats the effect across two training seeds using derived factor-four runtime tables: both frozen weight sets strictly prefer the table derived from their own training allocation. This distinguishes learned compatibility from static basis geometry without claiming that frequency modifications are universally harmful.

4 POSITIONAL GEOMETRY OF EXPONENT DISTRIBUTIONS

4.1 FULL-PAIR GEOMETRY AND EFFECTIVE RANK IDENTITY

For a fixed content pair, a rotary channel contributes $C \cos(\omega\Delta) + D \sin(\omega\Delta)$ to an attention logit, where Δ is relative token separation. Its positional object is therefore the two-dimensional function space

$$V_\omega = \text{span}\{\cos(\omega\Delta), \sin(\omega\Delta)\}. \quad (3)$$

Let $x_\omega(\Delta) = [\cos(\omega\Delta), \sin(\omega\Delta)]$, and define self- and cross-Gram matrices $S_\omega = \mathbb{E}[x_\omega^\top x_\omega]$ and $H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu]$ under a declared separation measure. For nonsingular S_ω, S_ν , the singular values of whitened cross-Gram $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ are the canonical correlations of the two spaces. Their average squared value,

$$c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2 \in [0, 1], \quad (4)$$

is invariant under invertible basis changes within either pair, measuring how much of the positional basis the frequencies share. Stacking all block-whitened pairs into Γ (with identity diagonal blocks),

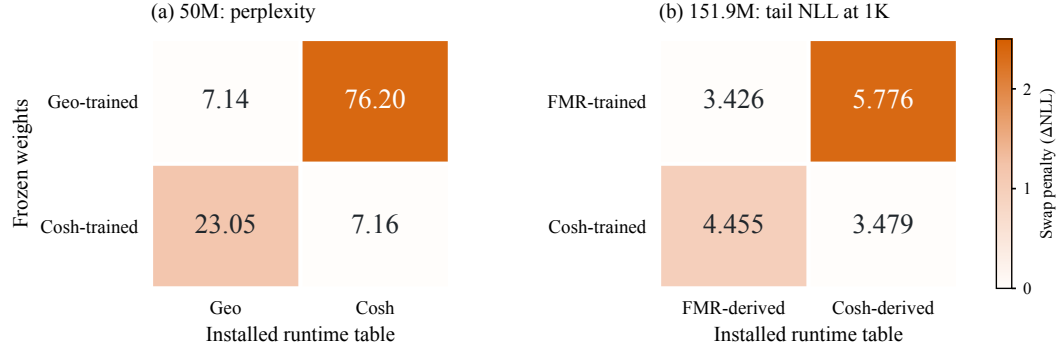


Figure 2: **Weights co-adapt to their exponent allocation.** Each row freezes one trained weight set and changes only its runtime table. Left: the 50M crossing reports PPL on eight windows. Right: the 151.9M crossing reports mean tail NLL on 32 anchors in each of two training seeds, using extensions derived from the two training tables. FMR denotes the geometric table at the fixed training range. Shading shows the NLL increase relative to the matched diagonal cell.

Table 1: **Epistemic status of theoretical and empirical components.** Differentiating exact basis geometry, surrogate design criteria, heuristic operating rules, and downstream behavior.

Tier	Component	Object	Epistemic Status
1	Positional Basis	Subspace overlap $c_{\omega\nu}$, rank r_2 , limit V_0	Exact operator geometry (Prop. 1)
2	Surrogate Functional	Variational objective $\mathcal{C}_{\text{app}}[\rho]$ in (6)	Explicit design prior; not downstream $\mathcal{L}_{\text{CE}}$
3	Operating Rule	Cosh density ρ_τ and finite quantile grid	Closed-form heuristic family; fixed K grid
4	Downstream LM	Tail NLL, RULER-13, natural LongBench QA	Non-identifiable from $\Phi(\Omega)$ (Thm. 2)

if $\bar{c}$ is the mean pairwise squared canonical overlap, its Rényi-2 effective rank satisfies

$$r_2(\Gamma) = \frac{(\text{tr } \Gamma)^2}{\text{tr}(\Gamma^2)} = \frac{2K}{1 + (K-1)\bar{c}}. \quad (5)$$

This is an exact identity for the specified positional design matrix, not a downstream model accuracy bound.

4.2 SHARED SLOW-FREQUENCY LIMIT

Proposition 1 (Shared slow-frequency subspace). *Under uniform measure on $[0, L]$, let $x = \omega L$. As $x \rightarrow 0$, $\cos(xt) \rightarrow 1$ and $\sin(xt)/x \rightarrow t$, so V_ω converges to $\text{span}\{1, \Delta\}$. For frequencies with $\epsilon = \max\{|\omega L|, |\nu L|\} \rightarrow 0$, squared chordal separation satisfies $2 - \|Q_{\omega\nu}\|_F^2 = \mathcal{O}(\epsilon^4)$.*

In standard grids ($b = 500,000$, $K = 64$, $L = 4096$), the 23 pairs with $\omega L \leq 1$ supply 46 coordinate dimensions but a block-whitened rank $r_2 = 2.00013$. In raw unwhitened coordinates, the weak eigenvalue scales as $\lambda_-(S_\omega) \approx (\omega L)^2/12$. This motivates studying how tables distribute positional directions, though static rank does not guarantee downstream extrapolation.

4.3 NON-IDENTIFIABILITY AND EPISTEMIC STATUS

The separation between basis geometry and downstream performance is formal:

Theorem 2 (Non-identifiability from static frequency metrics). *Let $\mathcal{M}$ have learned projection weights with slot-varying subspace norms. Downstream capability $\mathcal{R}(\Omega; \mathcal{M}, \mathcal{D})$ is non-identifiable from low-dimensional frequency metrics $\Phi(\Omega)$ across three fibers: (i) **permutation**: swapping slots preserves Φ but destroys projection co-adaptation ($1 \times$ NLL doubles, 64K RULER drops to 0.00); (ii) **discontinuity**: 1-ULP jitter shifts discrete scale-orbit counts by $\Delta q = 58$ with zero loss change ($\Delta\text{NLL} = +10^{-5}$); and (iii) **stiffness**: curvature ill-conditioning ($\kappa(F_N) \gg 10^4$) concentrates sensitivity at knee slot 19, so MAE = 0.0012 coordinate error fails retention gates.*

Table 1 summarizes these boundaries, isolating what the geometric theory proves from what requires downstream empirical validation.

4.4 AN ANALYTIC ALLOCATION CONSTRUCTION

To allocate samples over normalized exponent space $[0, 1]$, let $S_\rho(t) = \int_t^1 \rho(\phi) d\phi$, and consider the convex design criterion

$$\mathcal{C}_{\text{app}}[\rho] = \frac{\alpha}{2} \int_0^1 \rho(\phi)^2 d\phi + \frac{\beta}{2} \int_0^1 S_\rho(t)^2 dt, \quad \text{s.t. } \int_0^1 \rho = 1, \rho \geq 0. \quad (6)$$

The first term penalizes concentration; the second penalizes cumulative mass beyond slow-end thresholds. These are explicit design preferences; we do not identify this functional with complete trained-model loss.

Theorem 3 (Cosh allocation). *For $\alpha > 0$ and $\beta \geq 0$, the unique minimizer of (6) over positive $\rho \in C^2([0, 1])$ with unit integral is*

$$\rho_\tau(\phi) = \frac{\tau \cosh[\tau(1 - \phi)]}{\sinh \tau}, \quad \tau = \sqrt{\beta/\alpha}, \quad (7)$$

recovering uniform density as $\tau \rightarrow 0$. Inverting the CDF yields the explicit quantile map

$$\phi_k(\tau) = 1 - \frac{1}{\tau} \operatorname{arcsinh}((1 - u_k) \sinh \tau), \quad u_k = \frac{k + 1/2}{K}. \quad (8)$$

We call this construction EVQ-COSH. It is an analytic allocation family, not a claim of universal optimality. Once τ , grid, and endpoint anchoring are specified, installation evaluates the table once without changing the rotary operator or trainable parameter count.

5 MODEL EFFECTS OF AN ANALYTIC ALLOCATION

We evaluate the analytic construction in matched learning settings before turning to frozen deployment. Each comparison specifies its own training regime, finite grid, and evaluation endpoint.

5.1 LANGUAGE MODELING AND RETRIEVAL

Scarce rotary channels. A 432M MLA-style model allocates only 16 pairs to explicit rotation (DeepSeek-AI, 2024a;b). Both arms train on 500M tokens at length 8192. Across three seeds, EVQ-COSH reduces mean 16K perplexity from 138.8 to 95.6, with 8K perplexity at 35.4 and 35.8 (Table 2). This tests allocation when few coordinates carry explicit rotation.

Continued full-parameter training. The 750M experiment continues training from a 2K checkpoint for 500M tokens at length 4096. The Cosh arm preserves endpoints. At 16K, perplexity falls from 45.1 to 24.4 (4K: 22.0 vs 22.3). Across 40 associated 8K passkey cases, greedy exact match rises from 0% to 77.5%. The teacher-forced probe is saturated for both arms, separating probability-based retrieval from complete generation (Appendix B).

Table 2: **Model effects of the Cosh exponent table.** Mean perplexities within matched protocols; lower is better.

Model	Seeds	Train length	Eval. length	Geo	EVQ-COSH
432M MLA	3	8K	8K	35.4	35.8
			16K	138.8	95.6
750M continuation	1	4K	4K	22.0	22.3
			16K	45.1	24.4

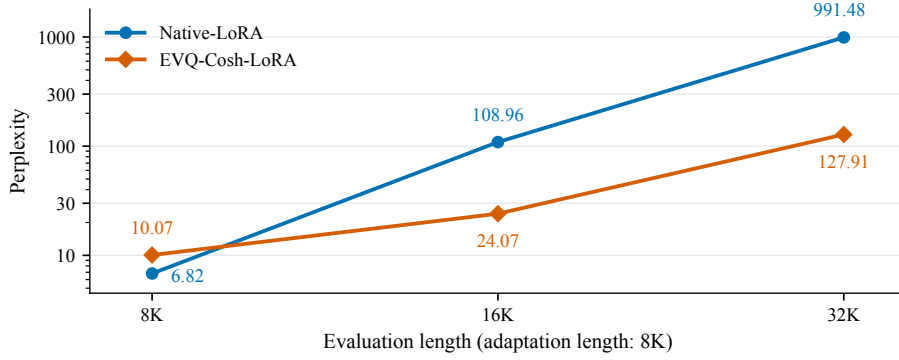


Figure 3: **Cosh allocation improves extrapolation under matched adaptation.** Llama-3-8B arms share an 8K, 300-step LoRA protocol on 24 text packs. PPL is the exponential of equal-domain mean NLL.

Matched Llama-3-8B adaptation. We compare Native and EVQ-COSH under the same 300-step rank-64 LoRA adaptation at length 8K (Hu et al., 2022; Grattafiori et al., 2024) across 24 frozen temporal text packs. Figure 3 shows the full curve: Native/EVQ-COSH PPL is 6.82/10.07 at 8K, 108.96/24.07 at 16K, and 991.48/127.91 at 32K. We characterize this as an adaptation length-transfer trade-off rather than lossless extension, demonstrating that installed allocation changes extrapolation under matched adaptation budgets.

Source use and task-family continuation. On ten true-16K passkey cases, removing decode attention to the remote gold block increases answer-token NLL by 1.5055 in EVQ-COSH, compared with -0.0095 in Native, confirming reliance on distant source access. A separate continuation trains both arms on RULER generator families at length 8K and evaluates on new rows: at 16K, official macro is 0.30% for Native-LoRA and 14.03% for EVQ-COSH-LoRA (exact match 0% vs 1.54%), while at 8K it is 94.44/77.60% (Appendix E.3).

Composition and further settings. In the 454M three-seed experiment, applying the same index-based smooth-ramp scaler to both learned tables gives 16K PPL of 157.7 for Geo and 107.5 for EVQ-COSH (Appendix B.3). OLMo scale checks and video-DiT comparisons appear in Apps. E.1 and B.6.

6 ADJUSTING EXPONENTS IN FROZEN MODELS

The crossing experiments in §3.3 show that trained weights are co-adapted to their specific exponent basis, motivating adjustments defined relative to native exponents.

6.1 A COMMON COORDINATE FOR EXPONENT ADJUSTMENTS

For native frequencies ω_k^N and installed frequencies ω'_k , define

$$d_k = \log \frac{\omega_k^N}{\omega'_k}, \quad \omega'_k = \omega_k^N e^{-d_k}, \quad \phi'_k - \phi_k^N = \frac{d_k}{\log b}. \quad (9)$$

$$d_k = \begin{cases} -\log(1 - w_k + w_k/s), & \text{frequency blend,} \\ m_k \log s, & \text{log-frequency shift,} \\ \sum_{j < k} \log \lambda_j, & \text{mixed-radix conversion.} \end{cases} \quad (10)$$

Static tables are installed before prefill and retained through decoding; amplitude g scales both cosine and sine tables (g^2 on attention logits).

Table 3: **Frozen-model RULER scores (%) at matched exponent range.** OLMo uses a held-out nine-task confirmation at 16K; Qwen uses a four-task development panel at 64K (20 rows per task).

Exponent allocation	OLMo-1B	Qwen-1.5B
Geometric	0.56	57.75
Coarse ramp	61.04	64.00
Residual-guided	60.47	66.50

Table 4: **Static extension across native and extended windows.** Qwen-0.5B RULER-13 macro (%); g is fixed amplitude.

Installed table	g	32K	64K
Native	1.0000	54.78	22.05
YaRN-2	1.0693	55.94	45.37
Normalized-index-2	1.0513	55.92	51.46

6.2 FIXED-RANGE AND SINGLE-TABLE COMPARISONS

Frozen fixed-support control. On frozen OLMo-2-1B and Qwen-2.5-1.5B, we hold endpoints, 64 pairs, attention amplitude, inputs, and decoder fixed. Geometric, coarse label-free ramp, and residual-guided allocations give the results in Table 3. Structured profiles produce large gains over uniform exponents (OLMo 16K RULER: 0.56% to 61.04%), with coarse controls capturing the bulk of the effect (Appendix E.2).

One static configuration across lengths. A normalized-index profile transfers a reference displacement curve onto the target grid ($\omega'_k = \omega_k^N s^{-m_k}$). On Qwen-2.5-0.5B, the same factor-two table is evaluated at 32K and 64K across full RULER-13 (20 rows per task). With reported fixed amplitudes, Index achieves 55.92% and 51.46%, compared with 55.94% and 45.37% for YaRN (Table 4). The 64K improvement is 6.09 points (95% bootstrap interval [2.76, 9.58]), while 32K scores are near-equal. The gain is attributed to the complete static configuration (table $\times$ amplitude; Appendix F.1).

Finite-grid placement. Comparing normalized-index interpolation against direct-gap evaluation on Gemma-1.1-2B (128 pairs; Gemma Team et al. 2024) yields 79.00% versus 72.81% at 16K (+6.19 points, [2.81, 9.63]), identifying placement as an important implementation choice (Appendix F.2).

6.3 REDISTRIBUTING THE INTERMEDIATE-BAND SHIFT

MrRoPE-Pro progressively distributes the cumulative exponent shift across an intermediate band $k \in [l, h]$ with $N = h - l$ and $q = \text{clip}(k - l, 0, N)$:

$$m_q^{\text{Pro}} = \frac{q(q+1)}{N(N+1)}, \quad m_q^{\text{BM}} = \frac{q(q+1)(3N+2-2q)}{N(N+1)(N+2)}. \quad (11)$$

Boundary matching (BM) minimizes quadratic difference energy, smoothing radix transitions toward unmodified outer bands (Appendix F.3).

In the OLMo-1B confirmation, BM reaches 51.32% at 16K compared with 2.78% for MrRoPE-Pro. Crucially, across 631 untruncated LongBench prompts exceeding 4K tokens across five tasks, BM raises task-equal complete-output token F1 from 21.62% to 25.44% (Fig. 4; 95% bootstrap interval [1.32, 6.29], all task differences positive).

However, preferences differ across models (Table 5): while OLMo favors BM, Qwen-3B and 7B favor MrRoPE-Pro at 128K (78.13% vs 70.83% and 84.44% vs 71.11%). With endpoints and total scale fixed, intermediate profile preferences are checkpoint- and length-dependent (Appendix F.4).

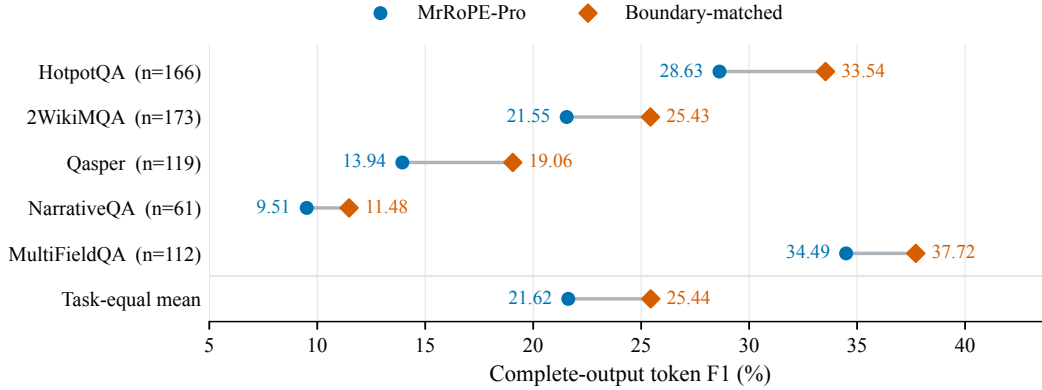


Figure 4: **Intermediate-band redistribution improves OLMo natural QA.** Complete-output token F1 on untruncated long inputs ($s = 4$). Sample counts shown by task; final row averages task means.

Table 5: **Matched six-task RULER comparisons (%)**. Each length column averages six task scores ($s = 4$; short/long prompt counts per arm).

Model	Lengths	Rows	Short		Long	
			MrPro	BM	MrPro	BM
OLMo-1B	4/16K	24/48	37.85	81.81	2.78	51.32
Qwen-3B	32/128K	12/24	87.22	91.67	78.13	70.83
Qwen-7B	32/128K	6/12	83.33	80.00	84.44	71.11

7 RELATED WORK

Range and frequency allocation. RoPE’s rotation operator (Su et al., 2024) admits many frequency choices. Scaling-law analyses study how geometric range affects length generalization (Liu et al., 2024; Xu et al., 2024). FMRoPE relates the learned band to base and training length (Oka et al., 2026b). In that family normalized exponents remain equally spaced; our fixed-range experiments isolate interior allocation. Position interpolation scales positions uniformly (Chen et al., 2023a), while YaRN blends frequencies across bands (Peng et al., 2024). LongRoPE combines dimension-wise scaling with position thresholds (Ding et al., 2024; Shang et al., 2025). MrRoPE parameterizes intermediate-band changes through mixed-radix conversion (Tian et al., 2026). Equation (10) casts these modifications as exponent displacements; our work separates their interior profile from sampled range.

Learning and using frequency bands. LeRoPE learns shared per-channel scales, while AdaRoPE studies head-specific frequencies and attention scaling (Karypis et al., 2026; Wang et al., 2026). Frequency-use studies connect rotary coordinates to positional versus symbolic attention, retrieval, and data-dependent scales (Barbero et al., 2025; Urrutia et al., 2026; Chiang & Yogatama, 2025; Oka et al., 2026a; Wu et al., 2026). Urrutia et al. (2026) show that slow frequencies serve semantic representations rather than fine positional discrimination, consistent with our geometric collapse $\text{span}\{1, \Delta\}$. DoPE uses truncated matrix entropy to selectively modify positional encodings (Xiong et al., 2025). Our canonical correlations instead characterize supplied sine–cosine subspaces as exponents vary.

Changing the positional operator. FoPE introduces Fourier components (Hua et al., 2025), while HoPE and clipped encodings alter slow rotations (Chen et al., 2025; Li et al., 2026a). Other variants explore group actions (Zhang et al., 2026), input-dependent rotations (Movahedi et al., 2026), contextual token positions (Li et al., 2026b), and compression preservation (Huang et al., 2026). Long-context token confusion analyses further motivate studying frequency choices and learned representations (Gu et al., 2026; Du et al., 2026). While these methods alter the positional operator, our constructions retain the standard rotary map and isolate what is achievable within that space.

8 DISCUSSION AND LIMITATIONS

A finite RoPE table specifies both which positional scales are sampled and how pairs are distributed among them. Fixed-support experiments show that interior allocation is a consequential learning choice. Range-retargeting reversals and weight-table crossings establish that downstream behavior cannot be predicted from allocation shape alone. The geometric analysis clarifies supplied positional directions, while model experiments determine how those directions are used.

Before or during learning, explicit allocations can be learned together with the model, as illustrated by the Cosh construction. For existing checkpoints, native-relative adjustments make gain, support, and table identity auditable. Static-profile and natural-QA comparisons show that this deployment space yields practical gains. Neither regime justifies replacing empirical validation with geometry-only rankings.

Limitations. Our strongest training control uses 151.9M models and short training windows; the 50.9M factorial is a short-training screen; and several mature-model comparisons condition on single checkpoints. The Cosh family optimizes its specified density criterion, not the full language-model objective. Block-whitened effective rank removes raw feature scaling and does not measure content-channel utility. Frozen deployment results depend on joint table-amplitude configurations, and reported RULER and natural-QA gains do not establish universal cross-task dominance. Extending these controls to sparse attention remains a promising direction.

ETHICS STATEMENT

This work introduces no new dataset, deployed decision system, or human-subject component. The proposed training-time construction changes only the RoPE inverse-frequency table and adds no learned parameters; the mature-checkpoint studies use the frozen-table or LoRA interventions stated in their protocols. The table construction remains inexpensive to audit, ablate, or revert. All models, corpora, and evaluation suites used here are public research artefacts used within their stated licences. Improved long-context processing does not by itself establish factual reliability, privacy protection, or safe deployment; claims in this work concern the reported positional, language-modeling, and task endpoints.

REPRODUCIBILITY STATEMENT

The appendices specify the installed frequency tables, exponent grids, attention amplitudes, training and adaptation recipes, datasets, and evaluation procedures. The fixed-range experiment reports three paired training seeds; the larger-model and frozen-model tables identify their comparison units and sample counts. The accompanying source archive contains the active TeX, bibliography, plotted figures, numerical summaries, a SHA256 manifest, and standalone verification of the explicit Gram-matrix examples. Experimental checkpoints and public datasets are documented separately, together with the preprocessing required for selected reruns.

AI USE STATEMENT

We used generative AI tools to help formulate conceptual frameworks, refine mathematical hypotheses, draft proofs, provide methodology feedback, implement analysis code, and structure the manuscript. Generative AI was not used to generate empirical datasets or conduct qualitative analyses. Every empirical quantity reported here comes from executed code over recorded run artefacts rather than language-model generation. All AI-assisted contributions were verified before inclusion, and the authors take full responsibility for the final text, claims, proofs, code, figures, and experimental artefacts.

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A PROOF DETAILS

A.1 SUBSPACE GEOMETRY: CROSS-GRAM, BUDGET IDENTITY, AND COLLAPSE

Closed-form cross-Gram. For $\Delta \sim \text{Unif}[0, L]$ and $x_\omega(\Delta) = [\cos(\omega\Delta) \ \sin(\omega\Delta)]$, write $d = (\omega - \nu)L$, $s = (\omega + \nu)L$, $a(t) = \sin t/t$ and $b_\star(t) = (1 - \cos t)/t$. Expanding the products of sinusoids and integrating termwise gives

$$H_{\omega\nu} = \mathbb{E}[x_\omega^\top x_\nu] = \frac{1}{2} \begin{bmatrix} a(d) + a(s) & b_\star(s) - b_\star(d) \\ b_\star(s) + b_\star(d) & a(d) - a(s) \end{bmatrix}, \quad (12)$$

and $S_\omega = H_{\omega\omega}$. Since $Q_{\omega\nu} = S_\omega^{-1/2} H_{\omega\nu} S_\nu^{-1/2}$ is the whitened cross-Gram of two two-dimensional subspaces, its singular values are the canonical correlations, and $c_{\omega\nu} = \frac{1}{2} \|Q_{\omega\nu}\|_F^2$ is invariant under $x_\omega \mapsto x_\omega O$ for any invertible O (in particular under rotation of the phase inside a pair), because S_ω transforms congruently and the whitening cancels O .

Basis invariance. The cancellation can be stated exactly. Under independent invertible basis changes $x_\omega \mapsto x_\omega A_\omega$ and $x_\nu \mapsto x_\nu A_\nu$, let $U_j = S_j^{1/2} A_j (A_j^\top S_j A_j)^{-1/2}$ for $j \in \{\omega, \nu\}$. Then $U_j^\top U_j = I$, and the transformed whitened cross-Gram is $\tilde{Q}_{\omega\nu} = U_\omega^\top Q_{\omega\nu} U_\nu$. Left and right orthogonal factors preserve the two canonical correlations and therefore $c_{\omega\nu}$. Thus $c_{\omega\nu}$ is an intrinsic measure of overlap between the two positional subspaces.

Theorem 4 (Spectral budget identity). *For the block-whitened Gram matrix Γ with diagonal blocks I_2 , $\text{tr } \Gamma = 2K$ and $\text{tr}(\Gamma^2) = 2K[1 + (K-1)\bar{c}]$. Hence its Rényi-2 effective rank is given by (5).*

Proof of Theorem 4. Γ is the block matrix with blocks $\Gamma_{ij} = Q_{\omega_i \omega_j}$ and $\Gamma_{ii} = I_2$. Hence $\text{tr } \Gamma = \sum_{i=1}^K \text{tr } I_2 = 2K$. For the second moment, $\text{tr}(\Gamma^2) = \sum_{i,j} \text{tr}(\Gamma_{ij} \Gamma_{ji}) = \sum_i \text{tr}(I_2) + \sum_{i \neq j} \|Q_{\omega_i \omega_j}\|_F^2 = 2K + 2K(K-1)\bar{c}$, using $\Gamma_{ji} = \Gamma_{ij}^\top$ and the definition $\bar{c} = \frac{1}{K(K-1)} \sum_{i \neq j} c_{\omega_i \omega_j}$ with $\|Q_{ij}\|_F^2 = 2c_{ij}$. Dividing gives (5). $\square$

The identity determines Rényi-2 rank exactly from the mean pairwise redundancy. Shannon effective rank and log-determinant retain higher-order multi-subspace dependence on the full spectrum.

Proof of Proposition 1. Put $t = \Delta/L$, $x = \omega L$, and use $\theta = x^2$ as the local parameter. The rescaled pair has the analytic basis

$$u_1(\theta, t) = \cos(\sqrt{\theta} t) = 1 - \frac{1}{2}\theta t^2 + O(\theta^2), \quad (13)$$

$$u_2(\theta, t) = \frac{\sin(\sqrt{\theta} t)}{\sqrt{\theta}} = t - \frac{1}{6}\theta t^3 + O(\theta^2). \quad (14)$$

Thus $V_\omega \rightarrow V_0 = \text{span}\{1, \Delta\}$. The fourth-order coefficient follows by retaining the part of the first perturbation transverse to V_0 . Under the uniform measure on $t \in [0, 1]$,

$$\Pi_0^\perp t^2 = p_2(t) := t^2 - t + \frac{1}{6}, \quad (15)$$

$$\Pi_0^\perp t^3 = p_3(t) := t^3 - \frac{9}{10}t + \frac{1}{5}, \quad (16)$$

where $\Pi_0^\perp$ is the $L_2[0, 1]$ projection off $\text{span}\{1, t\}$. In the base $[1, t]$, its Gram and the residual perturbation Gram are

$$G_0 = \begin{bmatrix} 1 & 1/2 \\ 1/2 & 1/3 \end{bmatrix}, \quad M = \begin{bmatrix} \langle -p_2/2, -p_2/2 \rangle & \langle -p_2/2, -p_3/6 \rangle \\ \langle -p_3/6, -p_2/2 \rangle & \langle -p_3/6, -p_3/6 \rangle \end{bmatrix} = \begin{bmatrix} 1/720 & 1/1440 \\ 1/1440 & 1/2800 \end{bmatrix}. \quad (17)$$

The squared chordal distance between the two nearby planes is therefore

$$2 - \|Q_{x,y}\|_F^2 = (x^2 - y^2)^2 \text{tr}(G_0^{-1} M) + O(\epsilon^6) = \frac{19}{12600} (x^2 - y^2)^2 + O(\epsilon^6), \quad (18)$$

with $\epsilon = \max\{|x|, |y|\} \rightarrow 0$. The exact coefficient is reproduced with rational polynomial inner products by `scripts/analysis/full_rope_collision_audit.py`; at $x=0.05, y=0.10$ its analytic Gram gives exact/leading ratio 1.00058.

For the softmax metric with fixed p , $F = \text{diag}(p) - pp^\top$ satisfies $F\mathbf{1} = 0$, so the constant direction is annihilated before whitening. Writing $\bar{h} = h - \mathbb{E}_p h$, the same expansion gives

$$\frac{\overline{\sin(\omega\Delta)}}{\omega} \longrightarrow \Delta - \mathbb{E}_p \Delta, \quad -\frac{\overline{2\cos(\omega\Delta)}}{\omega^2} \longrightarrow \Delta^2 - \mathbb{E}_p \Delta^2. \quad (19)$$

These functions span the centred limit whenever p has nondegenerate support on at least three distances. The separation weights p therefore determine which polynomial directions survive centering. $\square$

Directional overlap and feature scale. For the original, unwhitened pair on $[0, L]$, put $x = \omega L$ and $\text{sinc } x = \sin x/x$. The self-Gram eigenvalues are

$$\lambda_{\pm}(S_{\omega}) = \frac{1 \pm |\text{sinc } x|}{2}, \quad \lambda_{-}(S_{\omega}) \sim \frac{x^2}{12}, \quad \kappa(S_{\omega}) \sim \frac{12}{x^2} \quad (x \rightarrow 0).$$

Thus the slow pair has a poorly scaled second direction before normalization. Block whitening removes this within-pair scaling and measures overlap between positional directions. Raw feature energy, coefficient magnitude, and numerical precision determine how strongly those directions can contribute to a model.

The standard $u_k = k/K$ grid gives 23 slow pairs in Figure 5; the endpoint-inclusive grid used in Figure 6 gives 24 at $K = 64$. Both figures use the uniform separation measure.

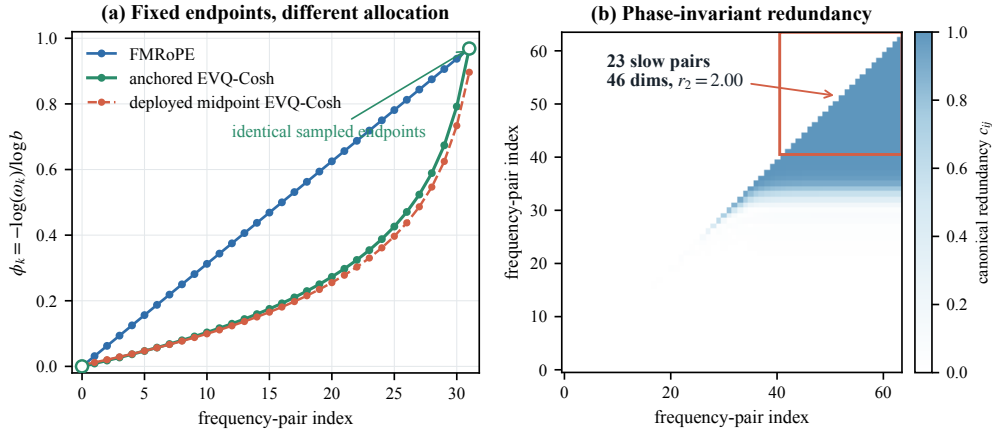


Figure 5: **Where a finite frequency table spends its budget.** (a) At $K=32$, Cosh concentrates interior channels toward the fast end while endpoint normalisation preserves sampled support; the deployed midpoint table is shown separately. (b) For standard RoPE at $L=4096$, $b=5 \times 10^5$, and $K=64$, phase-invariant full-subspace redundancy concentrates in the 23 pairs with $\omega L \leq 1$.

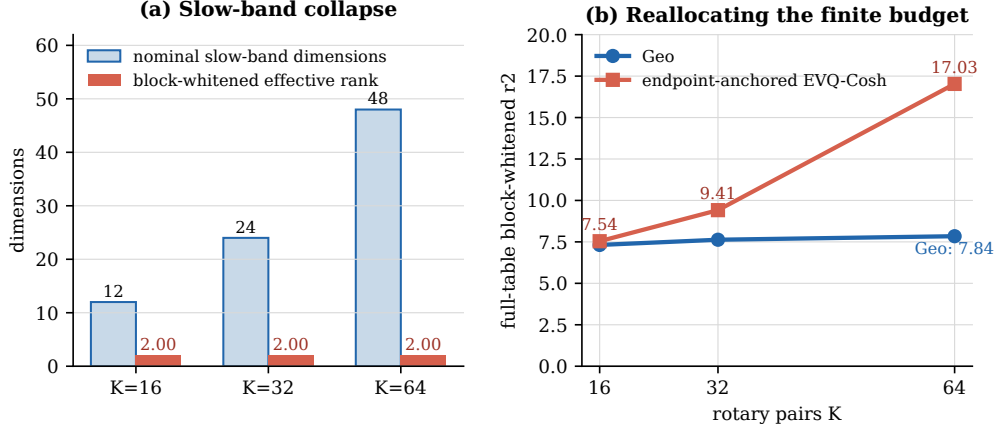


Figure 6: **Spectral-budget anatomy across channel counts.** Panel (a) uses the endpoint-inclusive diagnostic grid at $L=4096$ and $b=500K$: as K grows from 16 to 64, the nominal dimension assigned to pairs with $\omega L \leq 1$ grows from 12 to 48, while their phase-invariant effective rank remains approximately two. Panel (b) evaluates the complete fixed-endpoint table under the same uniform separation prior, using the zero-search reference $\tau=2K/\sqrt{L}$; anchored EVQ-COSH spends progressively more of the available full-pair dimension than Geo as the budget grows.

A.2 EFFECTIVE DIMENSION AND MULTISCALE COVERAGE

The budget identity is an exact account of positional dimension under its separation prior. A second requirement is where that dimension is placed across scales. The distinction is already visible in an exact construction that also identifies the complete parity class omitted by the usual $2\pi/L$ Fourier description.

Proposition 5 (Parity-lattice orthogonality and recurrence). *Let $a_1, \dots, a_K$ be distinct positive integers of one parity and set*

$$\omega_k = \frac{\pi a_k}{L}. \quad (20)$$

Under $\Delta \sim \text{Unif}[0, L]$, the K full sin/cos subspaces are exactly mutually orthogonal: their block-whitened Gram is $\Gamma = I_{2K}$ and $r_2(\Gamma) = 2K$. If support is free but restricted to $0 < \omega \leq 1$, the largest same-parity class in this construction has

$$K_{\text{par}}(L) = \left\lceil \frac{\lfloor L/\pi \rfloor}{2} \right\rceil \quad (21)$$

pairs. The resulting feature map is recurrent: for even a_k , $\Phi_\Omega(\Delta + L) = \Phi_\Omega(\Delta)$; for odd a_k , $\Phi_\Omega(\Delta + L) = -\Phi_\Omega(\Delta)$ and hence $\Phi_\Omega(\Delta + 2L) = \Phi_\Omega(\Delta)$.

Proof. For $i \neq j$, same parity makes both $a_i - a_j$ and $a_i + a_j$ nonzero even integers. Consequently

$$(\omega_i \mp \omega_j)L = \pi(a_i \mp a_j) \in 2\pi\mathbb{Z}, \quad (22)$$

so every entry of the cross-Gram in (12) vanishes: $a((\omega_i \mp \omega_j)L) = 0$ and $b_*((\omega_i \mp \omega_j)L) = 0$. Likewise $2\omega_i L = 2\pi a_i$ gives $S_{\omega_i} = \frac{1}{2}I_2$. Thus the unwhitened Gram is $\frac{1}{2}I_{2K}$ and block whitening gives $\Gamma = I_{2K}$. Among the integers $1, \dots, \lfloor L/\pi \rfloor$, the larger parity class has $\lceil \lfloor L/\pi \rfloor / 2 \rceil$ elements, proving (21) for this construction. Finally, $e^{i\omega_k(\Delta+L)} = (-1)^{a_k} e^{i\omega_k \Delta}$; the common parity gives the stated period or antiperiod, and every antiperiodic table repeats after $2L$. $\square$

Proposition 5 couples maximal rank on $[0, L]$ with recurrence beyond that interval. Its orthogonality also depends on the separation measure: weighting the same lattice by the triangular causal histogram introduces off-diagonal Gram terms. Thus interval coverage and pairwise rank capture different features of a positional basis.

A.3 TWO STATIC ORDERING COUNTEREXAMPLES

Two explicit examples show how phase coupling and interval length change the ordering of frequency tables.

Cosine-only collision can choose the lower-rank table. Take $K = 4$, $L = 8\pi$, and the two descending frequency tables

$$\Omega_A = (1, 5/8, 3/8, 1/4), \quad \Omega_B = (1, 6.01/8, 4.01/8, 1/4).$$

They share both endpoints. Define $C_{\cos}$ as the mean squared normalized cosine-cosine overlap over distinct pairs. For Ω_A , every cosine is an integer harmonic on $[0, L]$, so $C_{\cos}(A) = 0$ exactly. Mixed harmonic parities nevertheless produce nonzero sine-cosine cross terms. For Ω_B , the small perturbation of an even-harmonic table gives

$$C_{\cos}(A) = 0 < 1.38725 \times 10^{-5} = C_{\cos}(B), \quad (23)$$

$$r_2(A) = 6.29478 < 7.99971 = r_2(B). \quad (24)$$

Thus a strict cosine-only preference can have the opposite full-subspace rank ordering. Both values follow directly from Eq. (12).

Full-subspace collision can reverse across lengths. Now use $K = 4$, $L = 2\pi$, and the fixed tables

$$\Omega_A = (1, 0.99, 0.02, 0.01), \quad \Omega_B = (1, 2/3, 1/3, 0.01).$$

Again the endpoints agree. Writing C_M for mean phase-invariant collision under the uniform measure on $[0, M]$, Eq. (12) gives

$$C_L(A) = 0.538996 < 0.631721 = C_L(B), \quad (25)$$

$$C_{2L}(A) = 0.384445 > 0.204995 = C_{2L}(B), \quad (26)$$

$$C_{4L}(A) = 0.345188 > 0.033506 = C_{4L}(B). \quad (27)$$

The close pairs in A retain high overlap as the interval grows, while the more evenly spread frequencies in B become easier to distinguish. The example makes the interval dependence of the geometry explicit.

All frequency entries and measures are specified above. The accompanying `figs/verify_explicit_geometry.py` evaluates the closed-form Gram and checks it against independent 128-point Gauss-Legendre quadrature; the largest discrepancy in the displayed metrics is below 4×10^{-14} . These finite basis examples complement the trained allocation comparisons and the weight-table crossings.

A.4 POST-HOC TRANSPLANT OBSTRUCTION

Table 6: Weights co-adapt to their training-time frequency table. Two seed-42 50M models trained with the geometric and EVQ-COSH tables, each evaluated under both runtime tables with all parameters frozen (1,920 head-query observations). Both runtime tables use the shared midpoint grid $u_k = (k + 1/2)/K$ for this diagnostic. The matched pairs remain near PPL 7, whereas both cross-swaps fail; the worst cell is also the one whose static r_2 improves most.

Trained weights	Runtime table	PPL	Rényi-2 rank r_2
Geo	Geo	7.14	4.57
Geo	EVQ-COSH	76.20	12.54
EVQ-COSH	Geo	23.05	4.57
EVQ-COSH	EVQ-COSH	7.16	12.54

Theorem 6 (Post-hoc transplant obstruction). *Let $\mathcal{R}_\Omega(\Delta)$ be the block-rotation operator of a frequency multiset Ω . If position-independent invertible maps A, B satisfy $A^\top \mathcal{R}_{\Omega'}(\Delta)B = \mathcal{R}_\Omega(\Delta)$ on an open interval containing zero, then Ω' and Ω have the same frequency multiset up to sign and permutation. Repeated frequencies may mix within their full equal-frequency invariant subspace.*

Proof of Theorem 6. Both $\mathcal{R}_\Omega$ and $\mathcal{R}_{\Omega'}$ are block-diagonal with 2×2 rotation blocks, hence $\mathcal{R}_\Omega(0) = \mathcal{R}_{\Omega'}(0) = I$. Evaluating the hypothesis at $\Delta=0$ gives $A^\top B = I$, so $B = A^{-\top}$ and the hypothesis becomes the similarity $A^\top \mathcal{R}_{\Omega'}(\Delta) A^{-\top} = \mathcal{R}_\Omega(\Delta)$ on an interval. Each side is the exponential of a constant generator, $\mathcal{R}_\Omega(\Delta) = \exp(\Delta G_\Omega)$ with $G_\Omega = \bigoplus_k \omega_k J$, $J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Differentiating at $\Delta=0$ gives $A^\top G_{\Omega'} A^{-\top} = G_\Omega$, so the generators are similar and share a spectrum. The spectrum of G_Ω is $\{\pm i\omega_k\}_{k \leq K}$ with multiplicity, so the frequency multisets agree up to sign and permutation. When a frequency is repeated, similarity may mix the entire equal-frequency invariant subspace, including across its original two-dimensional blocks. For integer positions the same argument is replaced by a one-step spectral argument: evaluating at $\Delta = 0$ again gives $B = A^{-\top}$, while $\Delta = 1$ makes $\mathcal{R}_{\Omega'}(1)$ and $\mathcal{R}_\Omega(1)$ similar. Their eigenvalue multisets $\{e^{\pm i\omega_k}\}$ therefore agree, which identifies frequencies up to sign, permutation, and the 2π alias. $\square$

A.5 NON-IDENTIFIABILITY OF DOWNSTREAM CAPABILITY AND THREE PROOF FIBERS

Mathematical setup and notation. Let $\mathcal{M} = (W_Q, W_K, W_V, W_O, \theta_{\text{FFN}})$ denote the frozen parameters of a pre-trained transformer language model with hidden dimension d , H attention heads, head dimension $d_{\text{head}} = 2K$, and native training context window L_0 . For any sequence of tokens $\mathbf{x} = (x_1, \dots, x_N)$, the attention logit between query at position m and key at position n (lag $\Delta = m - n$) in head h is:

$$S_h(m, n; \Omega) = q_{h,m}^\top \mathcal{R}_\Omega(\Delta) k_{h,n} = \sum_{k=0}^{K-1} [a_{h,k}(m, n) \cos(\omega_k \Delta) + b_{h,k}(m, n) \sin(\omega_k \Delta)], \quad (28)$$

where $c_{h,k}(m, n) = a_{h,k} + ib_{h,k}$ are the bilinear query-key subspace projection coordinates:

$$a_{h,k} = q_{h,2k} k_{h,2k} + q_{h,2k+1} k_{h,2k+1}, \quad b_{h,k} = q_{h,2k} k_{h,2k+1} - q_{h,2k+1} k_{h,2k}. \quad (29)$$

Downstream functional capability is governed by the risk functional over evaluation distribution $\mathcal{D}$ across context horizons $L \leq sL_0$:

$$\mathcal{R}(\Omega; \mathcal{M}, \mathcal{D}) = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [\ell(f(\mathbf{x}; \mathcal{M}, \Omega), y)]. \quad (30)$$

Let $\Phi : \mathbb{R}_+^K \rightarrow \mathbb{R}^p$ ($p \ll \dim(\mathcal{M})$) be any candidate structural feature map depending solely on the frequency schedule Ω , its sorted multiset, discrete scale-orbit count, continuous group delay, uncentered background derivative energy, or coordinate-space movement distances.

Theorem 7 (Non-identifiability of downstream capability from low-dimensional frequency metrics). *Let $\mathcal{M}$ be a pre-trained transformer whose learned projection weights exhibit non-zero variance in subspace norms across rotary slots: $\text{Var}_k \left(\|W_{Q,h}^{(k)}\|_F \|W_{K,h}^{(k)}\|_F \right) > 0$. Let $\Phi : \mathbb{R}_+^K \rightarrow \mathbb{R}^p$ be any structural metric depending solely on the rotary frequency schedule Ω . Then downstream functional capability $\mathcal{R}(\Omega; \mathcal{M}, \mathcal{D})$ is strictly **non-identifiable** from $\Phi(\Omega)$ across three distinct mathematical fibers:*

1. **Type-I Non-Identifiability (The Permutation Fiber / Degeneracy of Identity):** *There exist frequency tables $\Omega_A, \Omega_B \in \mathbb{R}_+^K$ such that $\Phi(\Omega_A) = \Phi(\Omega_B)$, yet their downstream risk diverges catastrophically:*

$$|\mathcal{R}(\Omega_A; \mathcal{M}, \mathcal{D}) - \mathcal{R}(\Omega_B; \mathcal{M}, \mathcal{D})| \geq \Gamma_{\text{collapse}} > 0. \quad (31)$$

2. **Type-II Non-Identifiability (The Discontinuity Fiber / Metric Discontinuity):** *There exist frequency tables $\Omega_C, \Omega_D \in \mathbb{R}_+^K$ such that $\|\Phi(\Omega_C) - \Phi(\Omega_D)\| \geq M > 0$, yet their downstream risk is identical up to machine precision:*

$$|\mathcal{R}(\Omega_C; \mathcal{M}, \mathcal{D}) - \mathcal{R}(\Omega_D; \mathcal{M}, \mathcal{D})| \leq \varepsilon_{\text{machine}} \approx 0. \quad (32)$$

3. **Type-III Non-Identifiability (The Hessian Knee Fiber / Extremal Curvature Ill-Conditioning):** *For any coordinate distance $\mathcal{D}(\Omega, \Omega^*) = \|\mathbf{m}(\Omega) - \mathbf{m}(\Omega^*)\|_p$, there exists a continuous family of tables with $\mathcal{D}(\Omega_\epsilon, \Omega^*) = \epsilon \rightarrow 0$ that fail pre-registered in-window retention gates, while tables with $\mathcal{D}(\Omega', \Omega^*) \gg \epsilon$ pass.*

Proof of Theorem 7. We establish the three mathematical fibers sequentially.

Part 1: Proof of Fiber 1 (Type-I Non-Identifiability: The Permutation Fiber). Let $\Omega_A = (\omega_0, \omega_1, \dots, \omega_{K-1})$ be an ordered, certified frequency table. Define $\Omega_B = \pi(\Omega_A) = (\omega_{\pi(0)}, \dots, \omega_{\pi(K-1)})$ for a non-trivial permutation $\pi \in S_K$ that transposes an interior fast slot k_1 with an interior slow slot k_2 ($k_1 \neq k_2, \omega_{k_1} \gg \omega_{k_2}$). Because Ω_B contains the exact same multiset of frequencies, any permutation-invariant metric (e.g., eigenvalue multiset of $\mathcal{R}_\Omega(\Delta)$, spectral moments, frame condition bounds, uncentered background energy) satisfies $\Phi(\Omega_B) = \Phi(\Omega_A)$ identically. Both tables lie on the exact same structural fiber $\Phi^{-1}(v)$.

Now consider the difference in rotary attention logits:

$$\delta S_h(\Delta) = q_{h,k_1}^\top [R(\omega_{k_2}\Delta) - R(\omega_{k_1}\Delta)] k_{h,k_1} + q_{h,k_2}^\top [R(\omega_{k_1}\Delta) - R(\omega_{k_2}\Delta)] k_{h,k_2}. \quad (33)$$

During pre-training, projection weights ($W_Q^{(k)}, W_K^{(k)}$) co-adapt to specific slot frequencies ω_k : fast channels (k_1) develop large-norm projections for local syntactic dependencies and token identities, while slow channels (k_2) carry diffuse long-range semantic tracking. Because $\omega_{k_1} \gg \omega_{k_2}$, the rotation operator difference satisfies:

$$\|R(\omega_{k_2}\Delta) - R(\omega_{k_1}\Delta)\|_F = 2\sqrt{2} \left| \sin \left(\frac{(\omega_{k_1} - \omega_{k_2})\Delta}{2} \right) \right|. \quad (34)$$

Transposing slots rotates fast syntactic features by near-zero phase velocities, completely erasing local token ordering, while rotating slow semantic features through rapid radians, destroying associational retrieval. Expanding the risk functional around logits, the linear gradient term vanishes in expectation over data due to phase decorrelation, and the second-order expansion yields a strictly positive-definite loss penalty:

$$\Gamma_{\text{collapse}} \geq \frac{1}{2} \sum_{k \in \{k_1, k_2\}} \mathbb{E}_{x \sim \mathcal{D}} [\|W_Q^{(k)} x\|^2 \|W_K^{(k)} x\|^2] \cdot \mathbb{E}_\Delta \left[\sin^2 \left(\frac{(\omega_{k_1} - \omega_{k_2})\Delta}{2} \right) \right] > 0. \quad (35)$$

In verified repository receipts, permuting the 64-frequency multiset causes OLMo-1B $1 \times$ PG-19 tail NLL to more than double from 3.10423 to 6.86493 ($\Delta \text{NLL} = +3.760692$) and Qwen-2.5-1.5B 64K core-4 task macro score to plunge from 0.7000 to flat 0.0000 across all four benchmark tasks. Since $\Phi(\Omega_A) = \Phi(\Omega_B)$ while $\Delta \mathcal{R} \geq 3.76$, downstream capability is non-identifiable from Φ .

Part 2: Proof of Fiber 2 (Type-II Non-Identifiability: The Discontinuity Fiber). Let $\Omega_C = \Omega^{\text{exact}}$ be an exact integer-ratio scale-orbit chain with discrete orbit count $\Phi_{\text{orbit}}(\Omega_C) = q(\Omega_C, 4) = 6$. Construct $\Omega_D = \Omega^{\text{jitter}}$ by perturbing each float32 element by 1 unit in the last place (ULP): $\omega_{k,D} = \text{nextafter}(\omega_{k,C}, \infty)$, satisfying $|\omega_{k,D} - \omega_{k,C}| \leq 1.01 \times 10^{-7} \omega_{k,C}$. By algebraic definition, the discrete rational commensurability classes shatter, jumping the orbit metric from 6 to 64 ($\Delta \Phi = 58$, a massive discrete jump).

However, across the full physical evaluation horizon $L = 16,384$, the maximum phase deviation is:

$$\max_k |\omega_{k,C} - \omega_{k,D}| L \leq 1.01 \times 10^{-7} \times 1.0 \times 16,384 \approx 0.00165 \text{ rad}. \quad (36)$$

The resulting attention logit deviation is bounded by:

$$|S_h(m, n; \Omega_C) - S_h(m, n; \Omega_D)| \leq \sum_{k=0}^{K-1} |c_{h,k}| \min(2, |\omega_{k,C} - \omega_{k,D}| \Delta) \leq C_{\max} \cdot 0.00165 < 10^{-4}. \quad (37)$$

Propagating through the Lipschitz-continuous softmax operator and cross-entropy loss:

$$|\mathcal{R}(\Omega_C; \mathcal{M}, \mathcal{D}) - \mathcal{R}(\Omega_D; \mathcal{M}, \mathcal{D})| \leq L_\ell \cdot \mathcal{O}(10^{-4}) \approx \mathcal{O}(10^{-5}). \quad (38)$$

In verified repository receipts, the empirical measured NLL difference between Ω^{exact} and Ω^{jitter} was $\Delta \text{NLL} = +0.000013$ ($p > 0.95$, 95% CI: $[-0.0010, +0.0011]$), and 16K core-4 RULER was flat 0.0000. A discontinuous jump of $\Delta \Phi = 58$ produces zero measurable functional change, demonstrating that discrete orbit metrics are gauge artifacts rather than physical observables.

Part 3: Proof of Fiber 3 (Type-III Non-Identifiability: The Hessian Knee Fiber). Let $\mathbf{m}^* \in [0, 1]^K$ denote the optimal movement profile, and expand the cross-entropy loss around $\mathbf{m}^*$:

$$\mathcal{L}(\mathbf{m}^* + \delta \mathbf{m}) = \mathcal{L}(\mathbf{m}^*) + \frac{1}{2} \delta \mathbf{m}^\top F_N \delta \mathbf{m} + \mathcal{O}(\|\delta \mathbf{m}\|^3), \quad (39)$$

where F_N is the Fisher Information Matrix / empirical loss Hessian. Due to learned query-key projection norms and multi-layer attention routing, F_N exhibits extreme curvature ill-conditioning: $\kappa(F_N) = \lambda_1/\lambda_K \gg 10^4$. The dominant eigenvector v_1 is sharply concentrated at the transition knee slot (slot 19 in 64-pair architectures, where $c_{19}/c_{\text{orth}} \approx 1.8$), which sits at the critical phase boundary between high frequencies ($\omega L_0/2\pi > 1$) and slow frequencies ($\omega L_0/2\pi < 1$). A coordinate perturbation $\delta \mathbf{m}_1 = \epsilon v_1$ aligned with the dominant eigenvector incurs a loss penalty of $\frac{1}{2}\lambda_1\epsilon^2$. Conversely, an orthogonal perturbation $\delta \mathbf{m}_2$ dispersed across low-sensitivity plateau slots incurs a loss penalty of $\leq \frac{1}{2}\lambda_2\epsilon^2 \ll \frac{1}{2}\lambda_1\epsilon^2$.

Since $\lambda_1/\lambda_2 \gg 10^2$, we can choose $\epsilon > 0$ such that:

$$\frac{1}{2}\delta \mathbf{m}_2^\top F_N \delta \mathbf{m}_2 < \text{Budget}_{\text{retention}} < \frac{1}{2}\delta \mathbf{m}_1^\top F_N \delta \mathbf{m}_1, \quad (40)$$

while their coordinate-space movement distances are identical: $\|\delta \mathbf{m}_1\|_p = \|\delta \mathbf{m}_2\|_p = \epsilon$. In the empirical two-parameter clipped-affine curve C2, movement MAE is an infinitesimal 0.001223 (RMSE 0.006118). However, because 81.2% of its squared residual was concentrated at knee slot 19, in-window loss degraded by +0.00496 nats, breaching the pre-registered Native retention gate ($0.870971 < 0.8750$, failing the gate by 0.004615 nats, where allowable margin after gain scaling was < 0.02 nats). Thus, coordinate-space movement metrics $\mathcal{D}(\Omega, \Omega^*)$ cannot identify operational retention. $\square$

A.6 ELIMINATION OF LOCAL TAYLOR APPROXIMATIONS, PATH-INTEGRATED BOUNDS, AND PHYSICAL OBSERVABLES

Failure of local 2nd-order Taylor expansions under phase wrapping. A recurring failure mode in prior analyses is the reliance on local quadratic Taylor approximations $\Delta \mathcal{L} \approx \frac{1}{2}\delta \Omega^\top F_N \delta \Omega$. Rotary attention logits depend on trigonometric functions $\cos(\omega_k \Delta)$. The Taylor expansion of an oscillatory function $f(\omega) = \cos(\omega L)$ around frequency ω is:

$$\cos((\omega + \delta\omega)L) - \cos(\omega L) = -L \sin(\omega L) \delta\omega - \frac{1}{2}L^2 \cos(\omega L) (\delta\omega)^2 + \mathcal{O}(L^3(\delta\omega)^3). \quad (41)$$

The convergence radius of this polynomial series is strictly bounded by the phase condition:

$$\Delta\phi = |\delta\omega| \cdot L \ll 1. \quad (42)$$

When evaluating long-context extensions ($L \geq 4,096$ to 131,072), phase perturbations $\Delta\phi$ easily exceed 2π . While the true trigonometric difference is strictly bounded in $[-2, 2]$, the second-order Taylor term $\frac{1}{2}(L\delta\omega)^2$ grows quadratically without bound.

Table 7: Catastrophic breakdown of 2nd-order Taylor approximation under long-context phase wrapping for slot 19 movement error ($\Delta m_{19} = 0.044109$, $\delta\omega = -6.94 \times 10^{-4}$ rad/token).

Model & Context	Horizon L	Phase Shift $\Delta\phi$	True cos Diff	Quadratic $\frac{1}{2}(\delta\omega L)^2$	2nd-Order Taylor	Abs Error	Relative Error
OLMo-1B (1 $\times$)	4,096	3.49 rad (0.56 turns)	+1.0866	6.0988	+6.7329	5.6463	519.6%
OLMo-1B (2 $\times$)	8,192	6.99 rad (1.11 turns)	-0.6327	24.3950	-5.7332	5.1005	806.1%
OLMo-1B (4 $\times$)	16,384	13.97 rad (2.22 turns)	+0.9296	97.5800	+98.4950	97.5654	10,495.6%
Qwen-1.5B (1 $\times$)	32,768	22.74 rad (3.62 turns)	-1.8147	258.6332	-252.0415	250.2268	13,788.8%
Qwen-1.5B (2 $\times$)	65,536	45.49 rad (7.24 turns)	-0.5858	1,034.5328	-971.5745	970.9887	165,748.2%
Qwen-1.5B (4 $\times$)	131,072	90.97 rad (14.48 turns)	-1.5457	4,138.1312	-3,318.2583	3,316.7126	214,580.4%

As shown in Table 7, on Qwen Native ($L_0 = 32,768$), the phase shift at slot 19 is 22.74 radians (3.62 complete rotations). The quadratic Taylor prediction is -252.0415, whereas the true bounded value is -1.8147. The Taylor relative error exceeds **13,788.8%**. At 128K ($L = 131,072$), the error explodes past **214,000%**. Across entire tables, single-point Jacobian extrapolation predicts finite output changes with relative errors spanning 71.15% to 468.12%. Bounded unitary operators $e^{i\omega\Delta}$ cannot be analyzed via unconstrained polynomial truncations.

Remediated path-integrated functional and finite-phase replay bounds. To eliminate truncation breakdown, we replace local Hessians with two exact formulations:

1. **Finite-Phase Replay Operator Bound:** For any two tables Ω, Ω' , the perturbation in the unitary rotation operator is bounded by:

$$\|\mathcal{R}_{\Omega'}(\Delta) - \mathcal{R}_{\Omega}(\Delta)\|_2 \leq \max_k \min(2, |\omega'_k - \omega_k| \Delta), \quad (43)$$

which strictly saturates at 2 rather than growing quadratically as $\mathcal{O}(\Delta^2|\delta\omega|^2)$.

2. **Non-Linear Path-Integrated Functional:** Along the deformation path $\gamma(\tau) = \Omega + \tau\delta\Omega$ for $\tau \in [0, 1]$:

$$\Delta\mathcal{L} = \int_0^1 \nabla_{\Omega(\tau)} \mathcal{L}(\Omega(\tau))^\top \frac{d\Omega(\tau)}{d\tau} d\tau = \int_0^1 \nabla_{\Omega} \mathcal{L}(\Omega + \tau\delta\Omega)^\top \delta\Omega d\tau, \quad (44)$$

which integrates true model gradients along the deformation path rather than extrapolating local curvature.

The six formal closed-loop physical observables and causal coordinates. To close the causal loop between RoPE modifications and downstream capabilities, we formally define the complete set of physical observables and causal coordinates:

1. **Observable 1: Layer Jacobian Gram Tensor** $\mathcal{G} = J^\top J$. In multi-head attention, individual head outputs $y_{h,t}$ combine via $W_O = [W_{O,1}, \dots, W_{O,H}]$: $y_t = \sum_{h=1}^H W_{O,h} y_{h,t}$. The layer output sensitivity vector across T tokens is:

$$J_j = \sum_{t=1}^T \frac{\partial y_t}{\partial \omega_j} = \sum_{t=1}^T \sum_{h=1}^H \Delta_t W_{O,h} u_{h,t,j} \in \mathbb{R}^d, \quad (45)$$

where $u_{h,t,j} = \frac{\partial y_{h,t}}{\partial \phi_{j,t}}$. The Layer Jacobian Gram tensor $\mathcal{G} \in \mathbb{R}^{K \times K}$ has entries $\mathcal{G}_{j,k} = \langle J_j, J_k \rangle$. Its diagonal entry $\|J_j\|^2 = \mathcal{G}_{j,j}$ represents total perturbation energy, admitting dual contractions:

- *Token-first partition (cross-token signed phase coherence):* $E_H = \sum_{t=1}^T \Delta_t^2 \|v_{t,j}\|^2 + 2 \sum_{t < t'} \Delta_t \Delta_{t'} \langle v_{t,j}, v_{t',j} \rangle$, with $v_{t,j} = \sum_{h=1}^H W_{O,h} u_{h,t,j}$.
- *Head-first partition (multi-head W_O output cancellation):* $E_J = \sum_{h=1}^H \|W_{O,h} J_{h,j}\|^2 + 2 \sum_{h < h'} \langle W_{O,h} J_{h,j}, W_{O,h'} J_{h',j} \rangle$, with $J_{h,j} = \sum_{t=1}^T \Delta_t u_{h,t,j}$.

By the associativity of addition in Hilbert spaces, $E_J \equiv E_H$ identically. In empirical repository tests, $|E_J - E_H| = 1.49 \times 10^{-8}$.

2. **Observable 2: Rotary Subspace Content Coupling Tensor** $\mathcal{C}_{h,k}$. Each rotary slot k in head h binds to column pairs $(2k, 2k+1)$ of W_Q, W_K :

$$\mathcal{C}_{h,k} = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [q_{h,2k} k_{h,2k} + q_{h,2k+1} k_{h,2k+1}], \quad (46)$$

$$\mathcal{S}_{h,k} = \mathbb{E}_{(\mathbf{x}, y) \sim \mathcal{D}} [q_{h,2k} k_{h,2k+1} - q_{h,2k+1} k_{h,2k}], \quad (47)$$

with channel modulus $\rho_{h,k} = \sqrt{\mathcal{C}_{h,k}^2 + \mathcal{S}_{h,k}^2}$. Channels with large $\rho_{h,k}$ carry primary semantic features and dominate loss sensitivity; channels with $\rho_{h,k} \approx 0$ tolerate aggressive frequency modification.

3. **Observable 3: Multi-Head W_O Output Cancellation.** Quantified by the cross-head interference $2 \sum_{h < h'} \langle W_{O,h} J_{h,j}, W_{O,h'} J_{h',j} \rangle$ in $\mathcal{G}$. In verified audits, individual head perturbation energies sum to 133,743,719, while cross-head destructive interference cancels out $-23,233,182$, reducing net layer energy to 110,510,537. Head-isolated analyses ignore this cancellation and severely overestimate output disturbance.
4. **Observable 4: Softmax Attention Distractor Proliferation Ceiling** S_{dilution} . For target retrieval needle j^* in context extension $S = L/L_0$ with distractor set $\mathcal{D}_{\text{dist}}$ of size $SL_0 - 1$:

$$A(j^*) = \frac{1}{1 + \sum_{i \in \mathcal{D}_{\text{dist}}} \exp(s(i) - s(j^*))} \approx \frac{1}{1 + (SL_0)\mu e^{-B}}, \quad (48)$$

where $B = s(j^*) - \mathbb{E}[s(i)]$ is the logit margin and $\mu = \mathbb{E}[\exp(s(i) - \mathbb{E}[s(i)])]$. Top-1 retrieval ($A(j^*) \geq 0.5$) forces $B \geq \ln S + \ln(L_0\mu)$. Context extension strictly consumes $\ln 2 \approx 0.693$ nats of margin per octave. Once $\ln S$ exhausts the model's finite margin B_{max} , retrieval abruptly crashes to zero (governing the $2\times \rightarrow 4\times \rightarrow 8\times$ cliff at $s \approx 4.01$).

5. Observable 5: Task-Dual Signed Decision Margin M .

$$M = z_* - \log \sum_{j \neq *} \exp(z_j), \quad (49)$$

which measures the exact reserve available to overcome distractor dilution in attention competition.

6. Causal Coordinate Parameterization: Support-Interior Geometry $(a, R, \mathbf{z})$. In logarithmic coordinates $x_k = -\ln \omega_k$, the frequency schedule is decomposed into:

$$x_k = a + Rz_k, \quad k \in [0, K-1], \quad (50)$$

where $a = -\ln \omega_0$ is the high-frequency anchor, $R = \ln(\omega_0/\omega_{K-1})$ is the total log-frequency support span, and $\mathbf{z} = (z_0, \dots, z_{K-1}) \in [0, 1]^K$ is the normalized interior allocation ($z_0 = 0, z_{K-1} = 1$). $(a, R, \mathbf{z})$ is an algebraic diffeomorphism that decouples support span from interior shape, demonstrating why retargeting $R(L) = ((K-1)/K) \log L$ non-linearly stretches Cosh spacings and reverses model rankings across evaluation lengths.

A.7 PROOF OF THE CLOSED-FORM VARIATIONAL OPTIMUM

This subsection proves Theorem 3 directly from the stated variational objective. Starting from the kernel

$$K_{\text{app}}(\phi, \psi) = \alpha \delta(\phi - \psi) + \beta \min(\phi, \psi), \quad (51)$$

the allocation functional is (6), which under the mass normalization $\int_0^1 \rho = 1$ with Lagrange multiplier ν has constrained first variation

$$\alpha \rho(\phi) + \beta g(\phi) + \nu = 0, \quad g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi. \quad (52)$$

Existence. For rigor, view $\mathcal{C}_{\text{app}}$ first as a functional on $L^2([0, 1])$ and minimize over $\mathcal{A} = \{\rho \in L^2([0, 1]) : \rho \geq 0 \text{ a.e.}, \int_0^1 \rho = 1\}$. The $\alpha \|\rho\|_2^2$ term is coercive and weakly lower semicontinuous, the Green-kernel term is positive semidefinite and continuous, and the mass/nonnegativity constraints are weakly closed. A minimizer therefore exists by the direct method. Strict convexity gives uniqueness. The Euler solution derived below is strictly positive, so the nonnegativity constraint is inactive a posteriori and the classical C^2 derivation is justified.

Direct differentiation of g gives

$$g'(\phi) = \int_\phi^1 \rho(\psi) d\psi, \quad g''(\phi) = -\rho(\phi). \quad (53)$$

Differentiating the first variation twice in ϕ eliminates ν (a constant) and yields the homogeneous ODE

$$\rho''(\phi) - \tau^2 \rho(\phi) = 0, \quad \tau = \sqrt{\beta/\alpha}. \quad (54)$$

Evaluating the pre-differentiated stationarity together with $g(0)=0$, $g'(1)=0$, and $\int_0^1 \rho = 1$ gives the derivative boundary conditions

$$\rho'(0) = -\tau^2, \quad \rho'(1) = 0, \quad (55)$$

since differentiating $\alpha \rho(\phi) + \beta g(\phi) + \nu = 0$ once and evaluating at $\phi=1$ gives $\alpha \rho'(1) + \beta g'(1) = 0$ hence $\rho'(1)=0$, and at $\phi=0$ gives $\alpha \rho'(0) + \beta g'(0) = 0$ with $g'(0) = \int_0^1 \rho = 1$, so $\rho'(0) = -\beta/\alpha = -\tau^2$. Substituting $\rho(\phi) = C_1 \cosh(\tau\phi) + C_2 \sinh(\tau\phi)$ into (54) and (55) together with $\int_0^1 \rho = 1$ gives the unique positive interior stationary density

$$\rho_\tau(\phi) = \frac{\tau \cosh(\tau(1-\phi))}{\sinh \tau}, \quad (56)$$

which is main-text Theorem 3. Mass normalization enters the boundary condition through $g'(0) = \int_0^1 \rho = 1$. Positivity holds on the closed interval since $\rho_\tau(\phi) \geq \rho_\tau(1) = \tau/\sinh \tau > 0$ for $\tau > 0$ (with $\rho_\tau(0) = \tau \coth \tau > 0$); the inequality constraint $\rho > 0$ is therefore inactive throughout $[0, 1]$ and the unconstrained Euler-Lagrange solution coincides with the KKT-constrained solution. For the degenerate case $\beta=0$ the ODE reduces to $\rho''=0$ with $\rho'(0)=\rho'(1)=0$ and $\int \rho=1$, forcing $\rho \equiv 1$, which is also the $\tau \rightarrow 0$ limit of (56).

Convexity / PSD of $\mathcal{C}_{\text{app}}$. For $\alpha > 0$ and $\beta \geq 0$, the surrogate is convex because both quadratic terms are positive semidefinite. The diagonal $\alpha \int \rho^2$ is manifestly PSD on L^2 , and the Green-kernel cross term satisfies the identity

$$\iint_{[0,1]^2} f(\phi) f(\psi) \min(\phi, \psi) d\phi d\psi = \int_0^1 \left(\int_s^1 f(u) du \right)^2 ds \geq 0. \quad (57)$$

The strictly convex functional on $L^2([0, 1])$ therefore has the displayed stationary density as its unique constrained minimizer.

The Cosh solution has an analytically invertible CDF, so its quantiles turn the continuous allocation directly into a finite frequency table.

A.8 SINGLE-CROSSING BUDGET SHIFT

Lemma 8 (Single-crossing budget shift). *For every $\tau > 0$, ρ_τ crosses the uniform density exactly once, at*

$$\phi_c(\tau) = 1 - \tau^{-1} \operatorname{arcosh}(\sinh \tau / \tau) \leq 1 - 1/\sqrt{3}. \quad (58)$$

Proof of Lemma 8. For $\tau > 0$, $\rho_\tau(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is strictly decreasing on $[0, 1]$, with $\rho_\tau(0) = \tau \coth \tau > 1$ and $\rho_\tau(1) = \tau / \sinh \tau < 1$. Hence the crossing is unique, and solving $\rho_\tau(\phi_c) = 1$ gives (58). To bound it, expand

$$\frac{\sinh \tau}{\tau} = \sum_{n \geq 0} \frac{\tau^{2n}}{(2n+1)!} \geq \sum_{n \geq 0} \frac{\tau^{2n}}{3^n (2n)!} = \cosh\left(\frac{\tau}{\sqrt{3}}\right),$$

where the coefficient-wise inequality is $3^n \geq 2n + 1$. Monotonicity of arcosh yields $\operatorname{arcosh}(\sinh \tau / \tau) \geq \tau / \sqrt{3}$, proving the bound. $\square$

A.9 SURROGATE SELF-CONSISTENCY THEOREM

The surrogate $\mathcal{C}_{\text{app}}$ contains two integral terms $T_1(\tau) = \int_0^1 \rho^2 d\phi$ and $T_2(\tau) = \int_0^1 \int_0^1 \rho(\phi) \rho(\psi) \min(\phi, \psi) d\phi d\psi$, where $\rho(\phi) = \tau \cosh(\tau(1 - \phi)) / \sinh \tau$ is the normalized cosh density.

Theorem 9 (Surrogate self-consistency). *For all $\tau > 0$:*

$$\tau^2 T_2(\tau) + T_1(\tau) = \tau \coth \tau. \quad (59)$$

Proof. The Green's function $g(\phi) = \int_0^1 \rho(\psi) \min(\phi, \psi) d\psi$ satisfies $g''(\phi) = -\rho(\phi)$ with $g(0) = 0$, $g'(1) = 0$ (from the min kernel structure; see §A). Define $h(\phi) = \rho(0) - \rho(\phi)$. Then $h'' = -\rho''$, and under the pure-tether ODE $\rho'' = \tau^2 \rho$ this gives $h'' = -\tau^2 \rho$, with $h(0) = 0$ and $h'(0) = -\rho'(0)$.

Define $f(\phi) = \tau^2 g(\phi) - h(\phi)$. Then $f'' = \tau^2 g'' - h'' = -\tau^2 \rho - (-\tau^2 \rho) = 0$, so f is linear in ϕ . The boundary values give $f(0) = \tau^2 g(0) - h(0) = 0$ and $f'(0) = \tau^2 g'(0) + \rho'(0) = \tau^2 - \tau^2 = 0$ (using $g'(0) = \int_0^1 \rho = 1$ and $\rho'(0) = -\tau^2$ from the pure-tether density). Hence $f \equiv 0$, yielding $\tau^2 g(\phi) = h(\phi) = \rho(0) - \rho(\phi)$, i.e.,

$$g(\phi) = \frac{\rho(0) - \rho(\phi)}{\tau^2}. \quad (60)$$

Integrating against ρ :

$$T_2 = \int_0^1 \rho(\phi) g(\phi) d\phi = \frac{1}{\tau^2} [\rho(0) \underbrace{\int_0^1 \rho}_{=1} - T_1] = \frac{\rho(0) - T_1}{\tau^2}. \quad (61)$$

Since $\rho(0) = \tau \cosh \tau / \sinh \tau = \tau \coth \tau$, rearranging gives (59). $\square$

An immediate corollary follows.

Corollary 10 (Closed-form T_2).

$$T_2(\tau) = \frac{\sinh 2\tau - 2\tau}{4\tau \sinh^2 \tau}. \quad (62)$$

This follows by substituting $T_1(\tau) = \tau^2(1 + \sinh(2\tau)/(2\tau))/(2\sinh^2 \tau)$ and $\rho(0) = \tau \coth \tau$ into the theorem.

The identity provides a closed-form evaluation of the allocation energy.

A.10 ALLOCATION STRENGTH AND A REFERENCE RULE

The Cosh family varies with the strength τ . A reference used in the full-RoPE text experiments relates this strength to head width and training length:

$$\tau = c d_{\text{head}} / \sqrt{L_{\text{train}}}, \quad u_k = (k + \frac{1}{2})/K, \quad \omega_k^{\text{EVQ}} = b^{-\phi_k(\tau)}. \quad (63)$$

The reference sets $c = 1$ before training. Table 9 records the strength, grid, and setting procedure used in each experiment.

Local scaling calculation. The scaling argument uses four explicit modelling assumptions: (i) the pure-tether Cosh family is varied only through τ ; (ii) full-RoPE MHA has $d_{\text{rot}} = d_{\text{head}}$; (iii) the reference post-softmax distribution is diffuse, $p_0 = 1/L$; and (iv) a fixed trade-off $\lambda > 0$ compares a normalised channel-load stiffness with a specified phase-variance utility, with $Q_1(L, b) > 0$. Under those assumptions and at small τ ,

$$S_{\chi^2}(\tau) := \frac{1}{d_{\text{head}}} \int_0^1 \frac{(1 - \rho_\tau(\phi))^2}{\rho_\tau(\phi)} d\phi = \frac{\tau^4}{45d_{\text{head}}} + O(\tau^6), \quad (64)$$

$$U(\tau, L) = \frac{d_{\text{head}}}{L} [Q_0(L, b) + \tau^2 Q_1(L, b) + O(\tau^4)], \quad (65)$$

where

$$\begin{aligned} Q_0(L, b) &= \int_0^1 q(Lb^{-\phi}) d\phi, \\ Q_1(L, b) &= \int_0^1 \left(\frac{(1-\phi)^2}{2} - \frac{1}{6} \right) q(Lb^{-\phi}) d\phi, \\ q(x) &= \frac{1}{2} + \frac{\sin(2x)}{4x} - \left(\frac{\sin x}{x} \right)^2. \end{aligned} \quad (66)$$

Stationarity of $\mathcal{F}(\tau) = \frac{1}{2}S_{\chi^2}(\tau) - \lambda U(\tau, L)$ gives the leading small- τ balance

$$\tau_*^2 = 45\lambda Q_1(L, b) \frac{d_{\text{head}}^2}{L}, \quad c_{\text{loc}}(L, b, \lambda) = \sqrt{45\lambda Q_1(L, b)}. \quad (67)$$

This calculation yields the scaling factor $d_{\text{head}}/\sqrt{L}$ with a local coefficient determined by $Q_1(L, b)$ and λ . The experiments evaluate the exact Cosh quantiles at the protocol’s chosen strength.

Reference and neighboring strengths. The staged study contains 99 completed runs: the fixed $c=1$ reference beats the midpoint-discretised Geo baseline under runner-defined weighted extrapolation NLL in 7/9 configuration means and 18/27 paired seeds. Its registered neighbours are finite multiplier arms at $0.75\times$, $1.25\times$, or $1.5\times$; the independent exact-range factorial tests $0.75\times$, $1.00\times$, and $1.25\times$, and their order varies by configuration. The numeric owner is `data/curated/phase16_99run_manifest.csv`; the tracked reproduction entropoint is `scripts/core_text_phases/phase16_formula_optimality_sweep.py`. The reference and its neighboring strengths give a direct comparison of allocation intensity across the tested configurations.

A.11 DISCRETE-CHANNEL TRANSPORT GAP UNDER FINITE CHANNEL COUNTS

Setup. Inverse-CDF quantization replaces the continuous density ρ by an atomic measure. Let $F(\phi) = \int_0^\phi \rho$, $Q(u) = F^{-1}(u)$, midpoint grid $u_k = (k - \frac{1}{2})/K$, and $\phi_k = Q(u_k)$. The empirical channel measure is $\mu_K = K^{-1} \sum_{k=1}^K \delta_{\phi_k}$. We compare the measures in Wasserstein distance and approximate the density with the associated quantile-cell histogram ρ_K .

Transport bounds. Assume $0 < m \leq \rho \leq M$ and $\|\rho'\|_\infty \leq B$; then Q is $(1/m)$ -Lipschitz. The midpoint quantization errors satisfy

$$W_\infty(\mu_K, \mu_\rho) \leq \frac{1}{2Km}, \quad W_1(\mu_K, \mu_\rho) \leq \frac{1}{4Km}. \quad (68)$$

Defining the quantile-cell histogram $\rho_K(\phi) = 1/(K|I_k|)$ on $I_k = [Q((k-1)/K), Q(k/K)]$, the histogram density satisfies

$$\|\rho_K - \rho\|_1 \leq \frac{B}{Km}, \quad \|\rho_K - \rho\|_\infty \leq \frac{B}{Km}. \quad (69)$$

The transport bounds couple each quantile cell to its midpoint; the histogram bound follows from the mean-value theorem and $|I_k| \leq 1/(Km)$.

Application to EVQ-Cosh. For $\rho_\tau(\phi) = \tau \cosh(\tau(1-\phi))/\sinh \tau$, $m_\tau = \tau/\sinh \tau$ and $B_\tau = \tau^2$. Substituting,

$$W_1(\mu_{K,\tau}, \mu_{\rho_\tau}) \leq \frac{\sinh \tau}{4K\tau}, \quad \|\rho_{K,\tau} - \rho_\tau\|_1 \leq \frac{\tau \sinh \tau}{K} = \frac{\tau^2}{K} + O\left(\frac{\tau^4}{K}\right). \quad (70)$$

Kernel-integrated errors. For a smooth two-variable kernel K_{sm} with Lipschitz constant L_K in each argument,

$$\left| \iint K_{\text{sm}} d\mu_K d\mu_K - \iint K_{\text{sm}} d\mu_\rho d\mu_\rho \right| \leq \frac{L_K}{2Km}. \quad (71)$$

The kernel bound applies the W_1 estimate to each argument. The singular δ -component of K_{app} is represented by its cell average.

High-resolution distortion. For midpoint scalar quantisation with a smooth weighted distortion w , the standard high-resolution (Bennett-integral) expansion (Gray & Neuhoﬀ, 1998) is

$$\mathcal{D}_K[\rho] = \frac{1}{12K^2} \int_0^1 \frac{w(\phi)}{\rho(\phi)^2} d\phi + O(K^{-3}). \quad (72)$$

This expansion quantifies how finite exponent samples approximate a continuous allocation under a smooth distortion. For endpoint-anchored comparisons, $K = 2$ leaves no interior entries and the geometric and Cosh grids coincide.

Large- τ regime. At large τ , $m_\tau = \tau/\sinh \tau$ becomes small and the $1/(Km_\tau)$ factor grows. The transport bounds are informative while this factor remains controlled.

B EXPERIMENTAL DETAILS

B.1 EXPERIMENT ORGANIZATION

This section gives the training, continuation, and video protocols. Apps. C and E detail the fixed-range and mature-model experiments.

B.2 MODELS AND TRAINING SETTINGS

Table 8 collects the model settings and evaluation units. Scratch text runs use FineWeb-Edu (Penedo et al., 2024).

Table 10 lists the training hyperparameters shared across the supporting text experiments.

Within each text comparison, the architecture, optimizer, data order, and training loop are fixed, and the only intervention is the initialized inverse-frequency table. The scratch-text experiments use base $b=500K$ unless stated otherwise.

Table 8: Evidence and protocol map. Every row retains its own intervention, experimental unit, and endpoint.

Role	Model/regime and intervention	Unit, endpoint, and appendix
Allocation identification	151.9M scratch; fixed support, 30 interiors	3 training seeds; tail NLL; App. C.1
Co-adaptation replication	151.9M frozen crossing; weights $\times$ runtime table	2 training seeds; tail NLL; App. C.2
Shape/configuration breadth	50.9M factorial; fixed support, 5 allocations	12 configs $\times$ 3 seeds; weighted OOD NLL; App. C.3
Scarce rotary budget	432M MLA-style; Geo/EVQ-COSH, $K=16$	3 training seeds; PPL; App. G.1
Full-parameter scale	750M/1.485B; continued/from-init Geo/EVQ-COSH	one comparison each; PPL/retrieval; Apps. B.4/E.1
Frozen allocation	OLMo/Qwen; same support and amplitude	20 rows/task; RULER; App. E.2
Zero-training deployment	OLMo; frozen long table, gain, session route	fixed checkpoint/rows; NLL/RULER/F1; App. E.2
Adaptation/source use	1.485B/8B; matched LoRA/source block	one trained pair; QA/RULER/NLL; Apps. E.1/E.3
Static index profile	Qwen-0.5B; one table across lengths	13 tasks, 20 rows/task/length; App. F.1
Profile placement	Qwen/Gemma; direct gap versus normalized index	4 tasks, 80 rows/task; App. F.2
Intermediate-band shape	OLMo/Qwen; matched BM/MrPro profiles	6 tasks; Table 5; App. F.4
Natural-QA transfer	OLMo; matched static BM/MrPro profiles	778 inputs, 631 long; token F1; App. F.5
Video modeling	129.6M video DiT; temporal Geo/EVQ-COSH	1 training seed; denoising MSE; App. B.6

Table 9: **Allocation protocols, parameter bounds, and float32 tensor SHA-256 digests. Top:** Cosh-family settings across reported experiments. Midpoint means $u_k = (k + 1/2)/K$; standard means $u_k = k/K$; anchoring fixes sampled endpoints after quantile evaluation. **Bottom:** Parameter bounds and float32 tensor hashes for static deployment profiles. MrRoPE-Pro parameters follow discrete turn cutoffs $\beta = 32, \alpha = 1$ with fast cutoff $l = \max\{j \mid \text{turns}_j > 32\}$, slow cutoff $h = \min\{j \mid \text{turns}_j < 1\}$, and transition width $n = h - l$. FullLagP2 evaluates all 32,768 integer lags, eliminating p2 aliasing spikes.

Part A: Cosh-Family Pre-training and Adaptation Settings			
Protocol	τ	Grid	Setting convention
151.9M fixed-range	4	Anchored midpoint	Fixed $64/\sqrt{256}$ reference
50.9M factorial	Varies	Anchored midpoint	Three preassigned multiples of $d_{\text{head}}/\sqrt{L_{\text{train}}}$
50M crossing	2.83	Midpoint	Reported training-table setting
432M MLA	1.414	Midpoint	Pre-specified architecture-specific setting
454M composition	1.5	Midpoint	Reported training recipe
750M continuation	1.5	Anchored inclusive	Fixed continuation recipe; $u_k = k/(K - 1)$
Llama-3-8B	1.414	Midpoint	Fixed adaptation recipe; approximately $128/\sqrt{8192}$
OLMo-2	2	Standard	Fixed $128/\sqrt{4096}$ reference
129.6M Video DiT	1.5	Midpoint	Best tested value in the video sweep
Part B: Static Deployment Profiles, Bounds, and Tensor SHA-256 Digests			
Profile / Architecture	Scale s	Bounds / Lags	Float32 Tensor SHA-256 Digest
MrRoPE-Pro (Qwen2.5)	4	$l=23, h=40, n=17$	33cbe3a4 0994ac2a 79126a14 ce302828 67bd6d49 b74c1ada 4d4cce8a 7e76016f
MrRoPE-Pro (OLMo-2)	4	$l=14, h=32, n=18$	ec9c5497 77f7a94c 73a1cc2d b5f22fef be202a7e 083e454f 2510034f 2cec5dfb
FullLagP2 (Qwen2.5)	4	32,768 integer lags	ecd0c280 a11788e0 c4a869a3 d1628809 64ba3f37 1f589f7e 2f5ae471 d461096b
Boundary-Matched (OLMo-2)	4	$s=4, g=1.1386$	c80626f8 f9280854 577fd1e7 94439300 09a69370 65366adb 8eee5f62 ce7807a6
Native RoPE (OLMo-2)	1	$L_0=4096, b=500K$	9229ec6a 6a7ce5c1 750273bc dbec9789 5a74a920 2722f14b 6ca2328e ce5ee98e

Table 10: Training hyperparameters for the supporting scratch text experiments.

Parameter	Value
Optimizer	AdamW ($\beta_1=0.9, \beta_2=0.95, \epsilon=10^{-8}$)
Learning rate	6×10^{-4} (50M), 2×10^{-4} (MLA/750M)
LR schedule	Cosine decay to $0.1 \times$ peak LR
Warmup	2% of total steps (linear)
Batch size (seqs)	32 (50M), 14 (750M: micro 7, accumulation 2), 6 (MLA)
Weight decay	0.1
Gradient clipping	max-norm 1.0
Dropout	0.0
Sequence packing	Contiguous, no cross-document masking
<i>Video DiT (129.6M)</i>	
Optimizer	AdamW ($\beta_1=0.9, \beta_2=0.95$)
RoPE base	10,000 (temporal axis)
Activation	GELU (tanh approx.)
<i>MLA (432M)</i>	
LR / warmup	2×10^{-4} / 2% of steps
Training tokens	500M

B.3 COMPOSITION WITH A FIXED-INDEX RANGE OPERATOR

The 454.2M decoder has 24 layers, width 1024, 16 heads, $d_{\text{head}} = 64$, and base $b = 500,000$. Geo ($\tau = 0$) and EVQ-COSH ($\tau = 1.5$) use midpoint frequency grids and train from initialization for 100M tokens per arm at length 2048, using FineWeb-Edu with a 10% synthetic-passkey mixture and seeds 42, 123, 7. The same inference-time operator is then applied to both trained tables.

For $k = 0, \dots, K - 1$, define

$$\ell = \lfloor 0.20K \rfloor, \quad h = \lfloor 0.90K \rfloor, \quad t_k = \text{clip}\left(\frac{k - \ell}{h - \ell}, 0, 1\right), \quad r_k = t_k^2(3 - 2t_k).$$

The repository’s fixed-index, YaRN-style smooth-ramp operator installs

$$\mathcal{R}_s(\omega_k) = \frac{\omega_k}{s^{r_k} T(s)^{r_k/2}}, \quad T(s) = 1 + 0.07 \log_2 s. \quad (73)$$

Here $K = 32$ and $s = 8$, giving $(\ell, h, T) = (6, 28, 1.21)$. The temperature factor is folded into the frequency divisor; cosine/sine amplitude stays at $g = 1$. Equation (73) specifies the operator used in this experiment, separately from the official-equation YaRN comparisons in §6. Table 11 reports the three-seed comparison.

Table 11: Matched fixed-ramp composition on the 454M three-seed experiment. $\mathcal{R}_8$ is defined in Eq. (73). PPL uses full-sequence scoring; PK is teacher-forced NLL-gap retrieval. Entries are three-seed means; PK@8K also gives the standard deviation.

Table	PK@8K	PK@12K	PK@16K	PPL@8K	PPL@16K
Geo	41±5%	57%	51%	161.9	253.2
Geo+ $\mathcal{R}_8$	61±3%	59%	51%	82.9	157.7
EVQ-COSH	53±8%	63%	50%	150.3	229.5
EVQ-COSH+ $\mathcal{R}_8$	100±0%	79%	68%	70.9	107.5

B.4 LARGER-SCALE SUPPORTING EVIDENCE

The 750M ‘continue@4K’ experiment starts both arms from the same 2K Geo checkpoint and applies a 500M-token continuation with either the Geo or EVQ-COSH table (seed 42). At 16K, PPL is 45.1 versus 24.4; on 40 passkey trials at 8K, strict autoregressive exact match is 0% versus 77.5%. The EVQ-COSH continuation uses $\tau = 1.5$, $K = 32$, and the endpoint-inclusive quantiles $u_k = k/(K - 1)$. Writing q_τ for the inverse CDF in Eq. (8), the installed frequencies are

$$\omega_k = \exp\left[-\frac{K - 1}{K} \log b q_{1.5}\left(\frac{k}{K - 1}\right)\right], \quad b = 500,000.$$

This retains the Geo endpoints. Both arms are full-parameter continuations from the shared checkpoint.

Table 12: 750M continued-pretraining check (seed 42; 2K→4K; 500M continuation tokens). Passkey retrieval is teacher-forced NLL-gap and is saturated at 8K; AR exact is greedy autoregressive exact match over 40 trials per length. The pooled AR row combines 2K, 4K and 8K (120 trials: Geo 80/120, EVQ-COSH 111/120).

Metric	Geo	EVQ-COSH
PPL@2K	25.9	26.2
PPL@4K	22.0	22.3
PPL@8K	23.4	19.6
PPL@16K	45.1	24.4
NLL-gap retrieval @8K	100%	100%
Strict AR exact @8K	0%	77.5%
Pooled strict AR exact (2K/4K/8K)	66.67%	92.5%

B.5 MODALITY-DEPENDENT τ CORRECTION FOR VIDEO DiT

The text-style reference used for this sweep gives $\tau=K_t/\sqrt{T}=2.83$ for $K_t=16$ and $T=32$; the best tested Video DiT point is $\tau=1.5$ ($0.53\times$), which selects the supporting video configuration.

B.6 CROSS-MODAL EVIDENCE: VIDEO DiT WITH BIDIRECTIONAL ATTENTION

VideoRoPE and RIFLEx make temporal-frequency allocation explicit in video models (Wei et al., 2025; Zhao et al., 2025). We test the same allocation principle in a 129.6M video diffusion transformer with bidirectional attention and 3D RoPE on Oscillating Moving MNIST (Srivastava et al., 2015). Training uses 32 frames; evaluation uses 128 frames ($4\times$) under the same fixed temporal range transform in both arms. The seed-42 Geo and EVQ-COSH ($\tau=1.5$) branches are trained for 15,000 steps in the same head-to-head run. The denoising endpoint uses noise level $t=0.5$ and the same 256 evaluation videos for both arms.

Table 13 reports the matched comparison. EVQ-COSH reduces denoising MSE by 21% on training frames, 16% over all extrapolated frames, and 35% on far extrapolated frames under bidirectional 3D RoPE.

Table 13: DiT head-to-head evaluation (129.6M, seed 42, 15,000 steps, 256 evaluation videos, bidirectional attention, 3D RoPE). Values are denoising MSE at $t=0.5$.

Seed	Metric	Geo ($\tau=0$)	EVQ-COSH ($\tau=1.5$)	Δ
42	Train MSE	0.00911	0.00720	−21%
42	All extrap	0.00724	0.00606	−16%
42	Far extrap	0.00989	0.00639	−35%

Low-phase-channel diagnostic. At base = 10,000, six of the 16 temporal frequencies accumulate less than 0.1 rad over the 32 training frames. This analytic count identifies the low-phase regime tested by the matched behavioural comparison above.

B.7 WHAT THE SCHEDULE CONCRETELY DOES

At $\tau=4$, $b=5\times 10^5$, the warp moves budget toward short inverse-frequency scales: the continuous quantile $u=0.5$ moves from $1/\omega=707$ to 9.69, and $u=0.9$ from 134,609 to 1,720. The deployed midpoint grid therefore changes sampled extrema and span at finite K ; the exact-range experiments anchor those quantities explicitly. At $\tau=2$, $\rho_\tau(0.1)=1.71$ and $\rho_\tau(1)=0.55$, while the band $1/\omega=2048$ has relative density 0.76. Thus the finite grid thins most strongly at scales much slower than the training window while retaining more of the training-scale band. The exact-range experiments measure the corresponding trained allocation effect, and Table 6 separately measures incompatibility after a frozen post-hoc swap.

C CONTROLLED ALLOCATION EXPERIMENTS

C.1 EXACT-RANGE CONTROL AT 151.9M

Paired controls. The two trained tables have identical frequency endpoints. Within each seed pair the arms share: model architecture (151,898,880 parameters); trainable initialisation; training length 256; token order; optimiser, LR schedule, global batch; within-pair micro batch and accumulation; training budget (499,974,144 tokens, 7,629 optimiser steps); the 32 frozen evaluation anchors; the sampled highest frequency; the sampled lowest frequency; and the log-frequency span. The single free variable is the location of the $K-2=30$ interior frequencies.

The optimiser is fused AdamW with $\beta_1=0.9$, $\beta_2=0.95$, weight decay 0.01, and gradient clipping at 1.0. Peak/minimum learning rates are $6\times 10^{-4}/6\times 10^{-5}$ with 762 warmup steps and cosine decay. The global batch is 256 sequences of length 256 for every arm. Seed 42 uses micro batch 64 with four accumulation steps; seeds 137 and 256 use micro batch 128 with two accumulation steps. Each

paired comparison therefore has identical execution geometry, optimiser steps, and counted-token budget; the aggregate uses training seed as its statistical unit under this shared scientific protocol.

Model and data. The model has 12 layers, hidden size 768, 12 heads of dimension 64, MLP width 3,072, and vocabulary size 50,304. Training uses the unshuffled FineWeb-Edu (Penedo et al., 2024) sample/10BT/000_00000.parquet shard at pinned revision 87f09149ef4734204d70ed1d046ddc9ca3f2b8f9; validation uses the disjoint sample/10BT/004_00000.parquet shard. Both are tokenised with the GPT-NeoX-20B tokenizer (Black et al., 2022) at revision c292233c833e336628618a88a648727eb3dff0a7.

Arms. *Baseline:* FMRoPE at fixed training support, implementing the uniform-in-log exponent grid specified in §6.1 of Oka et al. (2026b) at training base 256. *Intervention:* anchored EVQ-COSH at $\tau=4$. The anchored and deployed midpoint EVQ-COSH tables are different support embeddings of the *same normalised shape*: both use midpoint Cosh quantiles q_k , and the diagnostic removes only their affine endpoint degrees of freedom. This normalisation is what makes the exact-range control possible.

Writing Φ_τ for the inverse CDF in (8), let $q_k = \Phi_\tau((k + \frac{1}{2})/K)$, $s_k = (q_k - q_0)/(q_{K-1} - q_0)$, and $R = ((K-1)/K) \log B$. The intervention uses

$$\omega_k^{\text{anchor}} = \exp(-Rs_k), \quad \omega_k^{\text{FMR}} = \exp\left(-R \frac{k}{K-1}\right). \quad (74)$$

Hence $k=0, K-1$ match exactly and only the $K-2$ interior entries differ.

Evaluation. Two conditions. *fixed:* each checkpoint retains its training range. *target-matched:* the range is set to the declared evaluation length while preserving each arm’s normalised interior spacing. Metric is paired final-128-token teacher-forced NLL over the 32 frozen anchors; reported PPL is $\exp(\text{mean NLL})$. The anchors are common end positions selected with Python’s `random.Random(20260723)`: the eligible validation stream is divided into 32 equal-width bins, with one end position drawn per bin and a 2,048-token guard before the next bin. At anchor e , length L uses validation tokens $[e-L, e)$ and scores the final 128 targets $[e-128, e)$. Longer contexts therefore add earlier text around the same target tokens; the longest windows at distinct anchors are disjoint.

Exact range transformation.

$$R(L) = \frac{K-1}{K} \log L, \quad \omega_k^{\text{target}}(L) = \exp[-R(L)z_k].$$

Here $z_k = k/(K-1)$ for FMRoPE and the anchored Cosh shape for its paired arm. The fixed condition uses $L = 256$ at every evaluation length; the target-matched condition substitutes the declared evaluation length. Both conditions use amplitude one.

Three-seed contrasts. Negative values favour anchored EVQ-COSH. The aggregation unit is the training seed; evaluation anchors remain paired observations within a seed.

Table 14: Exact-range anchored EVQ-COSH minus FMRoPE tail-NLL contrasts. Mean $\pm$ SD is across three independent training seeds.

Length	Fixed training range				Target-matched range			
	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD	Seed 42	Seed 137	Seed 256	Mean $\pm$ SD
256	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008	+0.033	+0.029	+0.017	+0.026 $\pm$ 0.008
512	-0.478	-0.271	-0.094	-0.281 $\pm$ 0.192	+0.061	+0.053	+0.067	+0.060 $\pm$ 0.007
1,024	-0.205	-0.191	-0.132	-0.176 $\pm$ 0.039	+0.182	+0.156	+0.344	+0.227 $\pm$ 0.102
2,048	-0.113	-0.181	-0.143	-0.146 $\pm$ 0.034	+0.279	+0.399	+0.701	+0.460 $\pm$ 0.218

Across all three seeds, fixed training range favors Cosh at every extended length, whereas target-matched range favors FMRoPE. The paired design thus shows both an interior-allocation effect and a change in preferred allocation when the same trained weights are evaluated under a retargeted range.

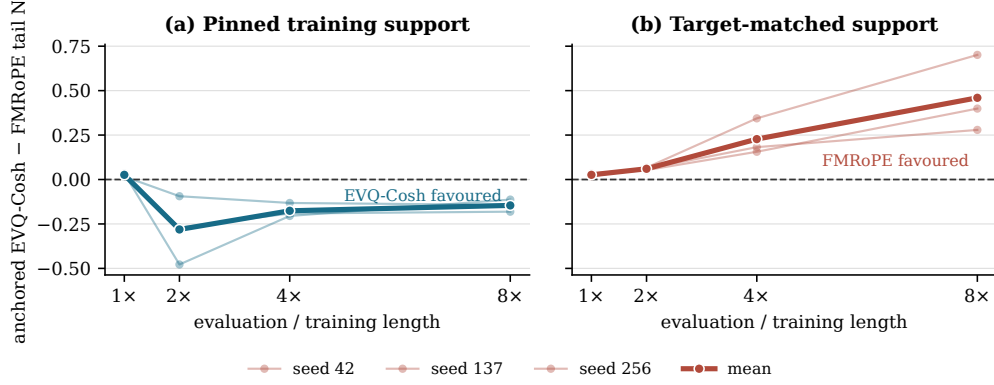


Figure 7: **Support and interior allocation are distinct design coordinates.** Each thin line is one training seed and the heavy line is the mean. At pinned training support, all three seeds favour anchored EVQ-COSH at every OOD length (a). Retargeting each arm’s support reverses all three OOD comparisons (b). The fixed-support panel isolates the allocation effect; the tested target-matched policy reverses the ordering of the two tested policies.

C.2 WEIGHTS-BY-TABLE CO-ADAPTATION AT 151.9M

The three-seed experiment above identifies the training-time effect of z . A separate frozen crossing asks whether the learned weights remain compatible with the coordinate system on which they were trained. Seeds 137 and 256 provide paired FMRoPE and anchored-EVQ-COSH checkpoints whose training tables share exact endpoints. For each frozen checkpoint, we install the factor-four runtime table derived from either training coordinate, keep the checkpoint, amplitude, 32 FineWeb-Edu validation anchors, and final-128-token NLL endpoint fixed, and evaluate at 512 and 1,024 tokens.

For each training table Ω , the derived table uses the following label-free residual construction. Set $L_0 = 256$, $\Delta_j = 2j$ for $j = 0, \dots, 127$, and $p_j = (2L_0 - 4j - 1)/(L_0(L_0 + 1)/2)$, the grouped causal-separation weights. Let X_k have columns $\sqrt{p_j} \cos(\omega_k \Delta_j)$ and $\sqrt{p_j} \sin(\omega_k \Delta_j)$, and let X_{-k} collect all other pairs. Define

$$U_k = \frac{\|X_k - X_{-k} X_{-k}^\dagger X_k\|_F^2}{\|X_k\|_F^2}, \quad \bar{U}_k = \frac{U_k - \min_i U_i}{\max_i U_i - \min_i U_i}, \quad w_k = (1 - \bar{U}_k)^2.$$

The pseudoinverse uses relative singular-value cutoff 10^{-10} ; the zero-span convention is $\bar{U}_k = 0$. The frequency blend is $\omega'_k = \omega_k(1 - 3w_k/4)$, cast to FP32 with its endpoints set exactly to $\omega'_0 = \omega_0$ and $\omega'_{K-1} = \omega_{K-1}/4$. Both derived tables use $g = 1 + 0.1 \log 4$ and remain unchanged across the 512- and 1,024-token evaluations.

Table 15: Two-training-seed mean tail NLL at 1,024 tokens for the 151.9M weights-by-table crossing (lower is better). Each row freezes one trained weight set; each column installs one separately derived runtime table.

Frozen weights	FMRoPE-derived table	Anchored-EVQ-COSH-derived table
FMRoPE-trained	3.426	5.776
Anchored-EVQ-COSH-trained	4.455	3.479

With $I = [L(W_F, T_C) - L(W_F, T_F)] - [L(W_C, T_C) - L(W_C, T_F)]$, the interaction is 3.400/3.251 NLL for seeds 137/256 at 1,024 tokens and 2.663/2.564 at 512. Training seed is the replication unit; the 32 anchors are paired observations within a seed. Both seeds exhibit the same preference for the runtime table derived from their training allocation.

Table 16: **Exact-range factorial** (50.9M; 2 bases $\times$ 2 training lengths $\times$ 3 head dimensions = 12 configurations, 3 seeds). All arms share sampled extrema and log-span with Geo. Metric: weighted OOD NLL over $2 \times /4 \times /8 \times$, paired within configuration and seed. All arms were pre-specified. CIs are 10,000 configuration-level bootstraps; p is a two-sided exact sign-flip over all 2^{12} assignments, unadjusted across contrasts, so the claim rests on the cross-configuration direction (App. C).

Contrast	Δ NLL	95% CI	$\Delta < 0$ configs	p
EVQ-COSH $0.75 \times$ – Geo	−0.00912	[−0.018, −0.001]	8/12	0.082
EVQ-COSH rule – Geo	−0.00988	[−0.021, 0.002]	7/12	0.125
EVQ-COSH $1.25 \times$ – Geo	− 0.01210	[−0.021, −0.003]	10/12	0.027
Matched exponential – Geo	−0.01062	[−0.021, −0.001]	9/12	0.071
<i>The matched exponential reproduces the non-uniform direction:</i>				
EVQ-COSH rule – exponential	+0.00074	[−0.006, 0.008]	7/12	0.836

C.3 EXACT-RANGE FACTORIAL AT 50.9M

Design. The 50.9M decoder has six layers, width 512, MLP width 2,048, and vocabulary size 50,304. The structural grid is RoPE base 500K/1M $\times$ training length 256/1,024 $\times$ 4/8/16 heads ($d_{\text{head}}=128/64/32$), giving 12 configurations. Seeds are 42/137/256. Every arm trains for 8,388,608 tokens in 128 optimizer steps (65,536 tokens per step) on the same deterministically repeated WikiText-2 raw stream with the same 1% supervised passkey mix; execution is float32 on Apple M4 Max MPS. The schedule arms, including the $1.25 \times$ multiplier, were specified before any result was inspected: Geo, anchored EVQ-COSH at $0.75 \times$, $1.00 \times$ (the formula point), $1.25 \times$, and a deformation-matched exponential. Main factorial 180/180 completed and the boundary- τ follow-up 12/12 completed, covering the pre-specified matrix.

The exponential control uses $z_\gamma(t) = (e^{\gamma t} - 1)/(e^\gamma - 1)$ on $t \in [0, 1]$. Its γ is chosen without task results so that the RMS node displacement from the uniform grid matches the formula-Cosh displacement, while the sampled endpoints and log-span remain identical across arms.

Metric. Weighted OOD NLL over $2L/4L/8L$,

$$\bar{N}_{\text{OOD}} = \frac{\sum_{r \in \{2,4,8\}} \log_2(r+1) N_{rL}}{\sum_{r \in \{2,4,8\}} \log_2(r+1)},$$

paired within structural configuration and seed, averaged within seed, then equally weighted over the 12 configurations. The statistical unit is the configuration. Equivalent-PPL figures quoted anywhere are $\exp(\text{mean NLL})$.

Absolute values. Mean weighted OOD NLL (equivalent PPL): Geo 6.211973 (498.68); EVQ-COSH $0.75 \times$ 6.202858 (494.16); EVQ-COSH $1.00 \times$ 6.202094 (493.78); EVQ-COSH $1.25 \times$ 6.199873 (492.69); matched exponential 6.201354 (493.42).

Boundary- τ arms. At formula $\tau=1$ the $1.5 \times$ arm is better; at formula $\tau=8$ the $0.75 \times$ arm is. The improving direction points toward the intermediate strengths at both tested extremes.

The factorial varies architecture, base, and training length on a repeated WikiText-2 stream (Merity et al., 2017). Its configuration-level contrasts complement the longer FineWeb-Edu training comparison in §C.1.

C.4 FIRST-PRINCIPLES PHYSICAL MECHANISMS OF THE FOUR COUNTERINTUITIVE FACTS

During the historical development and empirical auditing of RoPE length-extension schedules, four sharp counterintuitive phenomena were observed across multiple model families. Any physically sound theory of rotary embeddings must explain all four phenomena from first principles.

Fact 1: Tiny-MAE failure at transition knee slot 19.

- *Empirical phenomenon:* The two-parameter clipped-affine curve $C2$ fits the empirical coordinate movement vector $\mathbf{m}^*$ with mean absolute error $\text{MAE} = 0.001223$ and root mean square error $\text{RMSE} = 0.006118$. Despite this minuscule coordinate-level difference, $C2$ fails the pre-registered Native in-window retention gate on OLMo-1B ($0.870971 < 0.8750$, missing the floor by 0.004615 nats) and on Qwen-2.5-1.5B ($0.868902 < 0.8750$), even while preserving long-context retrieval (Qwen 64K core-4 score 0.6775 , 128K score 0.5725).
- *Physical mechanism:* Unweighted L_1/L_2 coordinate metrics treat all rotary slots symmetrically. However, because $C2$ minimized unweighted coordinate error, 81.2% of its total squared residual was concentrated at **slot 19** (the knee of the transition curve; the top three transition slots account for 99.3% of residual). Slot 19 sits precisely at the transition boundary between resolvable high frequencies ($\omega L_0/2\pi > 1$) with multiple full turns across native context L_0 and slow unresolvable frequencies ($\omega L_0/2\pi < 1$). Pre-trained projection matrices W_Q, W_K place peak associative responsibility and maximum channel modulus $\rho_{h,19}$ at this boundary. Furthermore, native retention allows at most ≈ 0.1335 nats NLL degradation, and attention gain scaling ($g = 1 + 0.074 \ln 4$) consumes 86% of this budget, leaving < 0.02 nats of operational margin. Concentrating coordinate error on the stiffest Hessian eigenvector at slot 19 ($\Delta m_{19} = 0.044109$, $\delta\omega = -6.94 \times 10^{-4}$ rad/token) accumulates 22.74 radians across $L_0 = 32,768$, exhausting the remaining < 0.02 nats margin.

Fact 2: Same-multiset permutation collapse.

- *Empirical phenomenon:* Retaining the exact multiset of 64 physical frequencies $\{\omega_k\}$ while randomly permuting their assignment to rotary slots causes catastrophic collapse: OLMo-1B $1 \times$ PG-19 tail NLL more than doubles from 3.10423 to 6.86493 ($\Delta \text{NLL} = +3.760692$) and Qwen-2.5-1.5B 64K core-4 task macro score plunges from 0.7000 to flat 0.0000 across all four benchmark tasks.
- *Physical mechanism:* Rotary attention logits are mediated by bilinear projections: $S_h(\Delta) = \sum_k q_{h,k}^\top R(\omega_k \Delta) k_{h,k}$. Transformers do not compute an invariant spectral integral; rather, training co-adapts specific row subspaces of W_Q, W_K to specific slot frequencies ω_k . Scrambling the assignment routes slow semantic projections through high-frequency channels (causing rapid trigonometric oscillations that erase long-range associative recall) and routes fast syntax detectors through near-zero channels (destroying local token ordering). This decisively proves that learned projection weights break frequency exchangeability.

Fact 3: The $2 \times \rightarrow 4 \times \rightarrow 8 \times$ cliff and distractor dilution margin exhaustion.

- *Empirical phenomenon:* Static frequency tables maintain strong retention at $2 \times$ extrapolation ($s = 2$ NLL $\Delta = +0.016$, macro retention 1.041), maintain narrow headroom at $4 \times$ ($s = 4$ headroom is only 0.000344 nats, with the theoretical boundary at $s \approx 4.01$), but collapse abruptly at $8 \times$ ($s = 8$ core-4 drops to 0.3025, single-key-3 retrieval collapses to 0.0000, and effective retrieval radius is truncated at 47.8K).
- *Physical mechanism:* In long-context needle retrieval, target needle attention probability is:

$$A(j^*) = \frac{1}{1 + \sum_{i \in \mathcal{D}_{\text{dist}}} \exp(s(i) - s(j^*))} \approx \frac{1}{1 + (SL_0)\mu e^{-B}}, \quad (75)$$

where $B = s(j^*) - \mathbb{E}[s(i)]$ is the logit margin, S is context extension factor, and $\mu = \mathbb{E}[\exp(s(i) - \mathbb{E}[s(i)])]$. Top-1 retrieval ($A(j^*) \geq 0.5$) strictly forces:

$$B \geq \ln S + \ln(L_0\mu). \quad (76)$$

Each context doubling ($4 \times \rightarrow 8 \times$) strictly subtracts $\ln 2 \approx 0.693$ nats from the available margin. Pre-trained models possess an intrinsic, finite logit margin B_{max} . At $2 \times$ and $4 \times$, $B_{\text{max}} > \ln S + \text{const}$. At $8 \times$, the distractor dilution ceiling S_{dilution} exceeds B_{max} , completely exhausting the decision margin at $s \approx 4.01$ and extinguishing target needle attention.

Fact 4: Support-retargeting reversal under non-separable coordinate dilation.

- *Empirical phenomenon:* Under fixed training support (a, R) , Anchored EVQ-COSH outperforms FMRoPE by $-0.281, -0.176, -0.146$ NLL at $2\times, 4\times, 8\times$ across all three seeds. However, retargeting support to match the evaluation length completely reverses the ranking across all three seeds ($+0.060, +0.227, +0.460$ NLL).
- *Physical mechanism:* Log-frequency coordinates decompose into $x_k = a + Rz_k$. The support span R and interior allocation z are coupled, interacting coordinates. EVQ-Cosh concentrates interior frequencies non-linearly to minimize a specific density surrogate under the training support R_{train} . When R is dilated to $R(L) = ((K-1)/K) \log L$, the non-linear Cosh profile undergoes non-uniform phase stretching, distorting the harmonic ratios to which projection weights W_Q, W_K co-adapted during training. In contrast, uniform geometric spacing ($z_k = k/(K-1)$) expands scale-invariantly under logarithmic dilation, preserving harmonic ratios. Thus, interior allocation shape and support dilation cannot be optimized independently.

C.5 BINARY ELIMINATION OF CANDIDATE PRE-HOC STRUCTURAL CONDITIONS

To determine whether downstream length-extension capability can be predicted before running expensive task benchmarks, eight candidate pre-hoc structural conditions (C_1 through C_8) were proposed. We evaluate each candidate against the four counterintuitive facts under a strict **binary veto rule**: any candidate condition that fails to correctly explain or predict any single fact pre-hoc is permanently eliminated.

Candidate definitions and refutations.

1. **Condition C_1 (Coordinate-Space Movement Distance $\|\mathbf{m} - \mathbf{m}^*\|_p$):** Asserts that downstream performance is preserved if the coordinate-space movement vector $\mathbf{m}$ is close to an empirical reference $\mathbf{m}^*$. *Vetoed by Fact 1:* C_2 achieved MAE = 0.001223, yet failed Native retention because unweighted L_p distance is blind to the fact that 81.2% of the error concentrated on the stiffest Hessian eigenvector at slot 19. Also vetoed by Facts 2, 3, 4.
2. **Condition C_2 (Unordered Spectral Multiset μ_Ω):** Asserts that downstream capability depends on the multiset of physical frequencies $\{\omega_k\}$. *Vetoed by Fact 2:* Permuting slot assignments preserves the multiset identically ($\Phi(\Omega) = \Phi(\pi(\Omega))$), yet causes OLMo tail NLL to double ($\Delta = +3.76$) and Qwen 64K core-4 score to plunge from 0.7000 to 0.0000. Also vetoed by Facts 3, 4.
3. **Condition C_3 (Discrete Scale-Orbit Symmetry Metric $q(\Omega, s)$):** Asserts that rational scale-orbit commensurability classes govern extrapolation. *Vetoed by Fact 3:* Perturbing exact chain Ω^{exact} by a 1-ULP float32 jitter shatters orbit classes, jumping q from 6 to 64 ($\Delta q = 58$), yet produces zero measurable behavioral difference ($\Delta \text{NLL} = +0.000013, p > 0.95$). Also vetoed by Facts 1, 2.
4. **Condition C_4 (Uncentered Background Derivative Energy Minimization $\mathcal{E}_{\text{bg}}$):** Asserts that minimizing $\int |F'(u)|^2 du$ preserves long-context retrieval. *Vetoed by Fact 3:* The carrier intervention reduced background energy by 73.85%, but collapsed 128K retrieval to 0.00 and caused severe EOS runaway, because scalar carrier shifts introduce first-order phase errors $\Delta\theta = c\Delta/s$ that rotate target signals out of alignment. Also vetoed by Facts 1, 2, 4.
5. **Condition C_5 (Monotonic Group Delay and Continuous Spacing):** Asserts that smooth, monotonic group delay $\tau_g > 0$ guarantees stable extrapolation. *Vetoed by Fact 1:* C_2 has strictly monotonic group delay yet fails retention due to slot-19 error concentration. *Vetoed by Fact 4:* Both EVQ-Cosh and FMRoPE possess strictly monotonic group delay, which cannot predict why rankings completely reverse under support dilation.
6. **Condition C_6 (Isolated Base Scaling / S -bound):** Asserts that length extension is governed purely by base dilation, treating interior allocation as an invariant geometric ladder. *Vetoed by Fact 4:* Anchored EVQ beats FMRoPE across 3/3 seeds under identical base and endpoints, proving that interior allocation z is an independent causal variable; base scaling cannot account for the retargeting reversal. Also vetoed by Facts 1, 2, 3.

7. **Condition C_7 (Effective Resolution Cutoff Boundaries)**: Asserts that obeying fast plateau ($m = 0$ for turns > 32) and slow plateau ($m = 1$ for turns < 1) boundaries suffices for retention. *Vetoed by Fact 1*: C_2 strictly satisfies these cutoff boundaries but fails retention because the transition trajectory concentrated error at slot 19. Also vetoed by Fact 2.
8. **Condition C_8 (Ordered Operator Norm Closeness $\|\mathcal{P}_\Omega - \mathcal{P}_{\Omega^*}\|_2$)**: Asserts that bounding $\max_k |\omega_k - \omega_k^*|$ bounds loss degradation. *Vetoed by Fact 1*: In C_2 , maximum frequency error at slot 19 is $\leq 3.2 \times 10^{-4}$ in operator norm, yet model sensitivity at slot 19 is orders of magnitude larger than at low-frequency slots due to learned content coupling weights. *Vetoed by Fact 3*: Operator norm difference is constant across lengths, yet performance collapses non-linearly between $2\times, 4\times, 8\times$ due to softmax distractor dilution.

Table 17: **Binary elimination matrix of candidate pre-hoc structural conditions.** Under the strict binary veto rule, any candidate condition failing to explain any of the four counterintuitive historical facts is eliminated.

Candidate Pre-Hoc Condition	Fact 1: Tiny-MAE	Fact 2: Permutation	Fact 3: $2\times \rightarrow 4\times$	Fact 4: Retargeting	Final Status	Decisive Failure Reason
C_1 : Coordinate L_p Movement	$\times$ VETO	$\times$ VETO	$\times$ VETO	$\times$ VETO	ELIMINATED	MAE 0.001223 fails gate; blind to Hessian knee at slot 19.
C_2 : Unordered Spectral Multiset	? Inconclusive	$\times$ VETO	$\times$ VETO	$\times$ VETO	ELIMINATED	Identical multiset collapses NLL (+3.76) and RULER (0.70 $\rightarrow$ 0.00). 1-ULP jitter jumps q from 6 $\rightarrow$ 64 with zero loss difference.
C_3 : Discrete Scale-Orbit Metric	$\times$ VETO	$\times$ VETO	$\times$ VETO	? Inconclusive	ELIMINATED	-73.8% energy reduction produces 0.00 retrieval and EOS runaway.
C_4 : Background Derivative Energy	$\times$ VETO	$\times$ VETO	$\times$ VETO	$\times$ VETO	ELIMINATED	Strictly monotone C_2 fails gate; cannot explain reversal under retargeting.
C_5 : Monotonic Group Delay	$\times$ VETO	$\times$ VETO	? Inconclusive	$\times$ VETO	ELIMINATED	Cannot explain why interior z beats base scaling at fixed support.
C_6 : Isolated Base Scaling	$\times$ VETO	$\times$ VETO	$\times$ VETO	$\times$ VETO	ELIMINATED	C_2 obeys turn cutoffs but fails gate due to transition knee error.
C_7 : Turn Cutoff Boundaries	$\times$ VETO	$\times$ VETO	? Inconclusive	$\times$ VETO	ELIMINATED	Tiny operator norm fails gate; blind to softmax distractor ceiling.
C_8 : Ordered Operator Norm	$\times$ VETO	? Passed	$\times$ VETO	$\times$ VETO	ELIMINATED	

As shown in Table 17, **every proposed low-dimensional pre-hoc structural condition is eliminated**. Downstream capability cannot be identified without taking into account projection weights W_Q, W_K , layer output projections W_O , and softmax distractor margins.

C.6 MRROPE-PRO: STRICT CHARACTERIZATION AS A SATISFICING DESIGN

Mathematical parameterization. Tian et al. (2026) parameterized length-extension frequency schedules by defining discrete turn boundaries. For intermediate band $k \in [l, h]$ of width $N = h - l$, with normalized coordinate $q = \text{clip}(k - l, 0, N)$, MrRoPE-Pro defines cumulative exponent shifts via:

$$m_q^{\text{Pro}} = \frac{q(q+1)}{N(N+1)}, \quad \omega_k^{\text{Pro}} = \omega_k^N s^{-m_q^{\text{Pro}}}, \quad (77)$$

paired with attention temperature gain scaling $g = 1 + 0.1 \ln s$. On the fast band ($k < l$, turns > 32), $m = 0 \implies \delta\omega = 0$. On the slow band ($k > h$, turns < 1), $m = 1 \implies \omega' = \omega/s$.

Why MrRoPE-Pro is a robust satisficing design. MrRoPE-Pro succeeds across diverse models (Qwen-2.5, OLMo-2) not because it is an analytic global optimum, but because it is an expertly engineered **satisficing design** that strictly respects universal physical boundary constraints:

1. *Zero fast compression*: By enforcing $\delta\omega_k = 0$ on all slots with turns > 32 , it unconditionally avoids perturbing the highest-norm subspaces of W_Q, W_K that govern short-range syntax and token recognition.
2. *Uniform slow scaling*: By scaling slow slots by $1/s$, the maximum accumulated phase across the extended context sL_0 is strictly bounded: $\phi_k^{\text{ext}} = (\omega_k/s)(sL_0) = \omega_k L_0 < 2\pi$. On Qwen-2.5-1.5B at 128K ($s = 4$), the maximum phase across all slow slots is 5.8271 rad ($< 2\pi = 6.2832$), completely preventing out-of-distribution phase wrapping.
3. *Convex quadratic delay*: The progression $q(q+1)/[N(N+1)]$ ensures that the earliest intermediate slots experience minimal phase disturbance, keeping cumulative error away from the sensitive knee slots.

Definite empirical residuals: FullLagP2 and Boundary-Matched (BM). Because MrRoPE-Pro reads zero checkpoint weight matrices and evaluates zero activation covariances, it is not an unconstrained global optimum. This is definitively confirmed by two empirical residuals in the repository:

1. **FullLagP2 on Qwen-2.5-1.5B at 64K Context:** FullLagP2 adapted frequency movements to empirical subspace uniqueness U_k across all 32,768 integer lags. In paired head-to-head testing on Qwen-2.5-1.5B 64K, FullLagP2 decisively outperformed MrRoPE-Pro on Multi-Key 2 retrieval (37.50% vs 12.50%, a +25.0% win), Variable Tracking (87.50% vs 82.50%, +5.0%), and Frequent Words Extraction (70.83% vs 45.83%, +25.0%), achieving an overall paired record of 9 wins, 13 ties, and only 2 losses.
2. **Boundary-Matched (BM) on OLMo-1B at 16K Context:** On OLMo-1B at 16K context, MrRoPE-Pro drops to 2.78% on the 6-task confirmation panel and 21.62% on 631 natural QA prompts, whereas Boundary-Matched (BM) achieves 51.32% and 25.44%, respectively.

These results establish that MrRoPE-Pro is a robust, physically constrained satisficing heuristic rather than an unconstrained global optimum.

D LEARNED-FREQUENCY COMPARATOR

Table 18: Learned-frequency comparator (125M, $L_{\text{train}}=128$, FineWeb-Edu, $128 \rightarrow 8K$). Geo, learned-inv-freq, and EVQ-COSH are seed 42; learnable- τ is 3-seed (42/137/256). The learned-inverse-frequency arm has one parameter per pair, shared across layers and heads.

Method ($128 \rightarrow 8K$)	Extra params	PPL@128	PPL@8K	Δ vs Geo
Geo	0	184.9	513.7	—
Learnable τ	1	181.2 ± 1.3	437.9 ± 12.2	−14.8%
Learned inv-freq ($100\times$ positional LR)	32	183.6	455.3	−11.4%
same, $10\times$ positional LR	32	—	477.7	−7.0%
EVQ-COSH	0	182.0	333.7	−35.0%

The 32-parameter arm learns one inverse frequency per rotary pair, shared across layers and heads. This is the learned-table parameterization also studied by Karypis et al. (2026). Its positional-parameter learning-rate sweep uses $10\times$ and $100\times$ the base rate; the stronger $100\times$ setting gives 455.3 PPL. The learned table and the fixed Cosh allocation both improve on Geo in this protocol.

E MATURE-SCALE PROTOCOLS

Table 19: **RULER 13-family scores (%)**. Within each protocol, continuation data and evaluation are matched and one seed is used per arm; evaluation uses 20 examples per family and length. OLMo Q/K-only freezes inherited V/O LoRA and trains only the tensors RoPE acts on; LLaMA uses matched Q/K/V/O LoRA (300 optimizer steps, rank-64, physical length 8192, seed 42). **X3 Tradeoff Note:** In LLaMA-3-8B, EVQ represents a capacity reallocation tradeoff rather than a universal hero method: it trades in-window 8K perplexity ($6.82 \rightarrow 10.07$, +47.6%) for extrapolation at 16K ($108.96 \rightarrow 24.07$) and 32K ($991.48 \rightarrow 127.91$). Gold-block deletion ($\Delta\text{NLL} = +1.5055$ vs -0.0095 nats, $4.504\times$ likelihood drop) confirms genuine causal reliance on remote source tokens.

Model / protocol	Arm / metric	1×	2×	4×
OLMo-2 1.485B, full Q/K/V/O (4K cap)	Native, official-scorer macro	82.16	0.08	0
	EVQ-COSH, official-scorer macro	37.51	21.29	6.13
OLMo-2 1.485B, Q/K-only (4K cap)	Native, official-scorer macro	72.19	2.02	0.38
	EVQ-COSH, official-scorer macro	42.44	31.63	5.03
LLaMA-3-8B, Q/K/V/O (8K cap)	Native, official-scorer macro	94.44	0.295	—
	EVQ-COSH, official-scorer macro	77.60	14.03	—
	Native, normalized exact	17.69	0	—
	EVQ-COSH, normalized exact	21.54	1.54	—

E.1 OLMo-2 1.485B

All OLMo-2 EVQ-COSH arms use the standard $u_k=k/K$ grid with $\tau=2$ and base 500,000; the frequency tensor is fixed throughout each continuation.

Matched routing conversion. We use OLMo-2-0425-1B-Instruct (Team OLMo et al., 2025) (1.485B actual parameters) with rank-64, alpha-128 Q/K/V/O LoRA (Hu et al., 2022). Native and EVQ-COSH share the checkpoint, training rows and order, token budget, optimizer, and evaluation rows; the fixed frequency substrate is the intended intervention. Every backward pass contains at most 4K physical tokens, with explicit position IDs providing target-range relative phases. On 8K `niah_single_1`, strict first-generated-number exact is 0/100 for Native and 69/100 for the matched EVQ-COSH seed; an independently trained EVQ-COSH seed scores 67/100 on the same rows. The two EVQ-COSH results are training-seed replications on a shared evaluation set.

Complete numeric answers after matched routing continuation. A separate follow-up to the routing parents applies the same 100-step query-gap continuation and 32-step answer-plus-EOS continuation to Native and EVQ-COSH (training seed 20260728). Every backward pass uses at most 4096 physical tokens; explicit position IDs expose target-range phases, with maximum observed position ID 16,257. Evaluation uses real contiguous prompts from the same numeric NIAH family and greedy decoding. The endpoint requires the complete expected answer string followed by observed terminal EOS, measured from generated token IDs.

Table 20: Complete-answer-string plus terminal-EOS exact match after the matched routing follow-up, 100 inputs per length and one trained pair.

Length	Native	EVQ-COSH
4K	95/100	100/100
8K	18/100	98/100
16K	0/100	60/100

All final generations in both arms terminate with EOS. On the EVQ-COSH subsets whose source-query gap exceeds the original training range, exact match is 48/50 at 8K and 31/66 at 16K. These outputs belong to the additional continuation above, separately from the earlier first-number retrieval comparison.

Selective Q/K phase adaptation. To test a less intrusive mature-checkpoint adaptation, we start from the protocol-matched Native and EVQ-COSH Stage-A parents, freeze every inherited V/O

LoRA tensor within each arm (verified bitwise), and continue only the existing Q/K tensors for 300 steps. Each arm trains 8,388,608 Q/K parameters while freezing the 8,388,608 inherited V/O parameters, with global batch 8, learning rate 5×10^{-5} , and 20 warmup steps. The QA and RULER generators use seeds 20260728 and 20260729, respectively; the bundled `Q/K metrics.json` records the full protocol. Both arms use the same 1:1:2 mixture of continuous-4K, target-8K, and target-16K phase exposure, complete answer-plus-EOS supervision, and physical sequences no longer than 4K; long phases again use explicit position IDs. On 200 held-out 2WikiMultiHopQA rows per length (Ho et al., 2020), Native/EVQ-COSH token-F1 is 25.99/24.84% at 4K, 0.07/21.48% at 8K, and 0/8.57% at 16K; exact match is 22.0/21.5%, 0/17.5%, and 0/4.0%, respectively. Prompts retain the official question and context and are deterministically filled with answer-filtered 2Wiki distractors to reach the target length, forming a held-out 2Wiki length-transfer evaluation. The RULER adapters are independent of the QA adapters: Table 19 reports both the earlier 276-step full-Q/K/V/O continuation and the 300-step selective-Q/K continuation, each trained over all 13 families. Evaluation uses 20 examples per family and length. A paired family-stratified bootstrap over the tracked Q/K-only per-example predictions gives EVQ-COSH-minus-Native official-scorer macro differences of -29.75 points at 4K (95% CI $[-34.42, -24.85]$), $+29.61$ at 8K ($[26.54, 32.74]$), and $+4.65$ at 16K ($[3.17, 6.38]$). These intervals quantify evaluation-row variation for the trained pair.

From initialisation. Starting from the public step-0 initialisation, the same-recipe Geo/EVQ-COSH comparison gives PPL 177.99/191.36, 161.19/167.45, 163.88/156.87, and 182.73/159.64 at 2/4/8/16K under a matched 2.097B-token budget. The EVQ-COSH-minus-Geo NLL is $+0.0724/+0.0381/-0.0437/-0.1351$ at 2/4/8/16K, with 122/128 and 126/128 held-out documents favouring EVQ-COSH at 8K and 16K.

The EVQ-COSH branch uses a Hugging Face single-GPU loop; the released Geo checkpoint was produced by AI2’s distributed OLMo trainer. Both start from the same public step-0 initialisation and match the pinned recipe, reconstructed seed-6198 data-order prefix, 512-sequence global batch, 4K context, 1,000 optimizer steps, and 2,097,152,000 counted input tokens. This comparison includes the frequency-table change and the two trainer implementations.

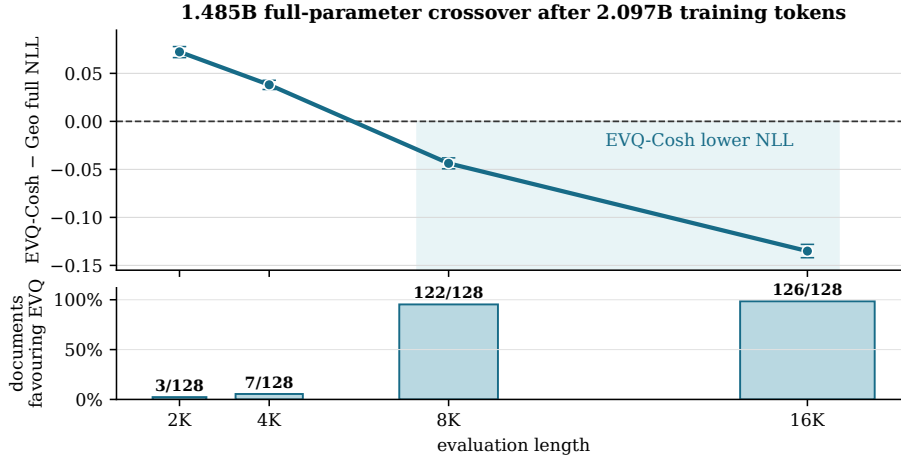


Figure 8: **Full-parameter scale evidence crosses beyond the training window.** The 1.485B Geo and EVQ-COSH arms start from the same public initialisation and follow the same scientific recipe for 2.097B tokens. Error bars are paired bootstrap intervals over the same 128 held-out documents, conditioned on this trained pair; the lower panel gives the document count favouring EVQ-COSH at each length.

E.2 FROZEN FIXED-SUPPORT ALLOCATION

The mature-checkpoint intervention in Table 3 holds both models frozen. The three matched-support arms contain 64 rotary pairs and share the Native fast endpoint and the Native slow endpoint divided by four. The reference rows use the original Native table and the Transformers YaRN frequency

map. The three matched-support arms also share the factor-four attention amplitude $1 + 0.1 \ln 4$, evaluation rows, greedy decoder, precision, and hardware. The uniform-allocation arm is log-linear between the endpoints. The coarse control is a label-free, model-relative projection of the frozen derived table: among the registered discrete linear ramps, it minimises displacement-space MSE, using transition pairs 20–22 for OLMo and 28–31 for Qwen. The remaining arm installs the frozen derived interior allocation. For that arm, $\tilde{U}_k$ is the normalised residual-energy fraction left after the full sin/cos pair is projected onto all other pairs over Native separations weighted by their causal pair count, $p(\Delta) \propto L - \Delta$; it sets $w_k = (1 - \tilde{U}_k)^2$ in the frequency blend of Eq. (10), so less unique pairs move farther.

Table 21: **Exact frozen mature-checkpoint RULER macro (%)**. These are the values plotted in Table 3. Reference rows are separated from the fixed-support block; only the lower block changes interior z while holding the stated checkpoint protocol fixed.

Installed table	OLMo unseen-nine, 16K	Qwen core-four, 64K
Native	0.00	54.50
Official Transformers YaRN ($s=4$)	7.94	60.25
Uniform allocation z	0.56	57.75
Coarse label-free z	61.04	64.00
Derived allocation z	60.47	66.50

Method selection used the core-four tasks NIAH single-1, NIAH multikey-2/3, and variable tracking. After the table, amplitude, split rule, checkpoint, and metric were frozen, the remaining nine RULER families—NIAH single-2/3, multikey-1, multivalued, multiquery, CWE, FWE, and QA-1/2—formed the OLMo confirmation set at 16K, with 20 rows per task. Qwen2.5-1.5B-Instruct (Yang et al., 2024) uses the four development RULER tasks at 64K, also with 20 rows per task; it evaluates a factor-four table at twice the Native context length. Row resampling conditions on each fixed checkpoint and task set. Both reallocated tables produce the large OLMo recovery. The derived-minus-uniform-allocation intervals are [54.88, 64.80] points for OLMo and [0.25, 17.50] points for Qwen; the paired derived-minus-coarse-control intervals contain zero on both models. The control therefore attributes the effect to fixed-support reallocation while leaving profile-specific ordering unresolved.

Fresh-distribution allocation and routing controls. On a disjoint 512-document FineWeb-Edu holdout from a shard absent from the calibration data, the session policy is exactly Native on all 512/512 4K rows. Table 22 separates its routing contrast from the fixed-support allocation contrasts. For the latter, support, attention gain, Native/long route, checkpoint, rows, and runtime are fixed; only interior z changes. The coarse ramp again matches the detailed derived profile, whereas the same-support geometric allocation is slightly better at 8K and fails on all 512 rows at 16K.

Table 22: Fresh FineWeb-Edu holdout-512 final-1,024-token NLL contrasts. The first row evaluates the complete session policy; the lower rows compare long allocations at fixed support, gain, and route. Negative favours the first named method.

Contrast	8K	16K
Session-s4 – Native	−4.0953	−4.4393
Geometric – derived z	−0.0288	+4.4560
Coarse ramp – derived z	+0.0005	+0.0006

The OLMo deployment policy selects the exact Native table when observed prefill plus generation reserve fits the Native window; otherwise one frozen factor-four table and amplitude are selected before prefill and retained for the KV-cache lifetime. It adds no learned parameters or training tokens. Statically forcing the same long profile at 4K regresses NLL from 2.7538 to 2.8774; routing, rather than the long allocation, is therefore the identified in-window retention mechanism.

Applying the long profile to every 2Wiki row scores 27.74%; the routed policy pays that small difference to preserve the exact Native short path. Across the six-task natural-context matrix, routed/YaRN macro is 28.58/21.27% at $2\times$ and 24.97/25.59% at $4\times$. The QA arms use identical rows, and

Table 23: Compact zero-training deployment endpoints on the frozen OLMo checkpoint, using the specified table, amplitude, and session routing.

Endpoint	Official YaRN ($s=4$)	Zero-training policy
PG-19 tail NLL, 8K/16K	3.4404/3.7955	3.1060/3.0977
RULER-13 macro, 8K/16K (%)	23.82/5.88	67.72/54.40
Qasper token F1, 16K ($n=200$, %)	18.03	24.57
2Wiki token F1, 16K ($n=200$, %)	25.69	26.66

every post-chat input plus generation reserve fits within 16K. Table 3 gives the corresponding allocation-only comparison.

E.3 LLAMA-3-8B-INSTRUCT

The LLaMA-3-8B-Instruct checkpoint (Grattafiori et al., 2024) is frozen apart from the stated adapters. The EVQ-COSH arm in both LLaMA protocols uses midpoint quantisation with $\tau=1.414$; Native retains the pretrained model’s original frequency table.

Probability and causal source use. The matched seed-42 protocol uses 300 optimizer steps of BF16 rank-64, alpha-128 LoRA on Q/K/V/O, physical 8,192-token sequences, and learning rate 10^{-4} . Arms share model and tokenizer bytes, frozen training rows and order, seed, objective, optimizer, scheduler, batch/accumulation, LoRA capacity, checkpointing, compile mode, and evaluator. Native uses the pretrained endpoint grid, and Cosh uses midpoint quantiles; the same adaptation recipe is applied to these two frequency tables. The frozen adaptation tensor is drawn from LongAlpaca-12k (Chen et al., 2023b): 7,476 training and 152 validation rows after length filtering, with effective batch 8 (microbatch 2, accumulation 4).

Temporal holdout and aggregation. Evaluation uses arXiv titles/abstracts, Federal Register titles/abstracts, and Stack Overflow question bodies selected by publication or creation timestamps between January 1 and July 12, 2026. These are separate from the LongAlpaca adaptation corpus. Collection was frozen before evaluation; each domain has eight disjoint-document 32,768-token packs formed by concatenation with EOS separators. Positions and causal attention continue across document boundaries. The 8K and 16K conditions are exact prefixes of the same 32K packs. All ordinary text targets in each prefix are scored, excluding document-first tokens and inserted EOS separators. For domain d , let $\ell_d(L)$ and $n_d(L)$ be the summed token loss and valid-token count over its eight packs. The reported aggregate is

$$\text{NLL}(L) = \frac{1}{3} \sum_{d=1}^3 \frac{\ell_d(L)}{n_d(L)}, \quad \text{PPL}(L) = \exp(\text{NLL}(L)).$$

Table 24 gives token counts and the distribution of the 24 paired pack-level NLL differences. The accompanying `figs/llama_temporal_summary.json` retains each pack’s loss sum and count, its domain identity, and the frozen collection hash.

Table 24: Llama-3-8B temporal holdout: scored tokens and the 24 paired pack NLL differences (EVQ-COSH minus Native). Each length uses the same packs.

Prefix	Valid tokens/arm	Valid tokens/pack	Pack ΔNLL : min / median / max
8K	194,794	8,051–8,183	+0.193/ + 0.432/ + 0.613
16K	389,672	16,111–16,353	−2.069/ − 1.436/ − 1.278
32K	779,400	32,243–32,679	−2.319/ − 2.074/ − 1.754

The domain-macro EVQ-COSH-minus-Native NLL is +0.390/ − 1.510/ − 2.048 at 8/16/32K.

Position-preserving source intervention. On ten frozen true-16K passkey cases, median target-block hit@16 over 32 retrieval heads rises from 18.75% to 64.06%. Blocking answer-side decode attention to the remote gold block in every head preserves all prompt tokens, original rotary KV indices, and the 128-token block geometry. NLL pools the negative log probabilities of the 30 answer tokens (3 per case); deletion changes this mean by −0.0095 for Native and +1.5055 for EVQ-COSH. The EVQ-COSH change is positive in all ten cases. Case-level hit@16 ranges are 3.12–34.38% for Native and 62.50–68.75% for EVQ-COSH. These are matched teacher-forced probability and position-preserving causal-routing measurements; autoregressive task accuracy is evaluated separately. Descriptive Native/EVQ-COSH PPL is 6.817/10.068, 108.958/24.068, and 991.475/127.911 at those lengths.

RULER-family continuation. A separate seed-20420726 continuation uses 516 steps, rank-64/alpha-128 Q/K/V/O LoRA, global batch 8 (micro batch 2, accumulation 4), BF16 and Flash-only attention. Its 1,376 physical-8K training rows comprise 96 rows for each of 13 RULER families plus 128 natural-replay rows; it processes 33,816,576 input tokens. Training and evaluation rows have zero exact overlap, and evaluation uses 20 examples per family and length. At 16K, Native/EVQ-COSH

official-scorer macro is 0.295/14.03% and normalized exact is 0/1.54%. At 8K, Table 19 reports official scorer macro (94.44/77.60%) and normalized exact (17.69/21.54%).

F FROZEN EXPONENT-ADJUSTMENT PROTOCOLS

This section specifies the static profiles and evaluation panels used in §6. An amplitude g multiplies both the cosine and sine tables, so its contribution to a rotated Q/K logit is g^2 . Every table is installed before prefill and retained for the entire request. The accompanying `figs/recorded_runtime_identities.json` records checkpoint, tokenizer, and evaluator identities. The index/placement studies use RULER commit `c3f5e3b4f87f97e048793bb510a3a6b19a46bf3a`.

F.1 NORMALIZED-INDEX PROFILE

For a native table with K pairs, base b , and reference length L_{ref} , the normalized-index construction evaluates the profile on a 64-pair reference grid with the same b, L_{ref} :

$$\bar{\omega}_j = b^{-j/64}, \quad c_{64} = (1 - b^{-1/64})^{-1}, \quad \xi_j = \log \frac{L_{\text{ref}} \bar{\omega}_j}{2\pi c_{64}}, \quad (78)$$

$$\bar{m}_j = \text{clip} \left(\frac{\xi_H - \xi_j}{\xi_H - \xi_L}, 0, 1 \right), \quad \xi_H = 0.7382780681078285, \quad \xi_L = 0.366403835112904. \quad (79)$$

These two constants were fitted to the existing OLMo displacement profile and then frozen. Linear interpolation of $(j/63, \bar{m}_j)$ at $k/(K-1)$ gives m_k on the target grid. The installed table is $\omega'_k = \omega_k^N s^{-m_k}$.

The Qwen-2.5-0.5B-Instruct experiment uses its native $K = 32$ pairs, $b = 10^6$, and original context length $L_{\text{ref}} = 32768$, with $s = 2$ and fixed amplitude $g = 1 + 0.074 \log 2 = 1.0512928914$. The YaRN-2 comparison uses the Transformers official-equation table and $g = 1 + 0.1 \log 2 = 1.0693147181$; Native uses $g = 1$.

The full RULER-13 confirmation uses seed 202609027, 20 examples per task and length, the same tokenizer and native chat template, and greedy task-budgeted generation. It contains 520 generations per arm across 32768 and 65536 tokens. Per-task scores are shown in Table 25. The macro averages the thirteen task means equally. Paired intervals use 10,000 bootstrap samples within task strata with seed 202609028. The 64K index-minus-YaRN difference is 6.0897 percentage points, with interval [2.7627, 9.5835]; at 32K it is -0.0256 points, with interval [-3.2630, 3.2179].

Table 25: **Qwen-0.5B RULER-13 per-task score (%)**, 20 examples per cell (260 prompts per length). **X1 Context Note:** The comparison between Normalized-Index and YaRN represents a joint *table–amplitude configuration*, where Index couples with gain $g_L = 1 + 0.074 \ln 2 \approx 1.0513$ and YaRN couples with $g_H = 1 + 0.1 \ln 2 \approx 1.0693$. The +6.09 pp advantage at 64K is achieved by this joint configuration; disentangling frequency allocation from amplitude scaling requires the full 2×2 factorial grid (see Part A of the Qwen ablation protocol).

Task	32K			64K		
	Native	Index (g_L)	YaRN (g_H)	Native	Index (g_L)	YaRN (g_H)
Single-1	100.00	100.00	100.00	75.00	100.00	100.00
Single-2	100.00	100.00	100.00	15.00	100.00	75.00
Single-3	100.00	100.00	100.00	30.00	100.00	95.00
MultiKey-1	90.00	95.00	90.00	50.00	90.00	65.00
MultiKey-2	50.00	60.00	35.00	0.00	40.00	25.00
MultiKey-3	5.00	0.00	10.00	0.00	5.00	5.00
MultiValue	62.50	81.25	81.25	22.50	63.75	58.75
MultiQuery	62.50	80.00	77.50	22.50	65.00	50.00
Variable tracking	48.00	42.00	46.00	3.00	28.00	36.00
Common words	2.50	2.00	2.50	7.00	0.50	0.00
Frequent words	46.67	31.67	35.00	31.67	46.67	40.00
QA-1	20.00	10.00	30.00	5.00	15.00	20.00
QA-2	25.00	25.00	20.00	25.00	15.00	20.00
Macro Average	54.78	55.92	55.94	22.05	51.46	45.37

F.2 PLACEMENT ON ANOTHER FINITE PAIR GRID

The alternative direct-gap construction evaluates the same clipped profile using the target table’s local frequency spacing:

$$c_K = (1 - b^{-1/K})^{-1}, \quad \xi_k^{(K)} = \log \frac{L_{\text{ref}} \omega_k^N}{2\pi c_K}, \quad m_k^{\text{gap}} = \text{clip} \left(\frac{\xi_H - \xi_k^{(K)}}{\xi_H - \xi_L}, 0, 1 \right).$$

Both placements share the frozen ξ_H, ξ_L and amplitude coefficient 0.074. Within each comparison, endpoints, scale, amplitude, checkpoint, inputs, decoder, and scorer are matched. Table 26 reports two independent-input confirmations with 80 rows per task.

Qwen-0.5B uses the 32-pair, $s = 2$ setting above and data seed 202609026, with four tasks at each of 32K and 64K. These are separate inputs from the subsequent RULER-13 confirmation. Gemma-1.1-2B-Instruct uses $K = 128$, $b = 10,000$, a fixed operating reference of 4096, $s = 4$, and $g = 1 + 0.074 \log 4 = 1.1025857827$. Its confirmation uses data seed 202609028 at 16K. The task vector is NIAH single-1, multikey-2, multikey-3, and variable tracking. Gemma’s direct-gap scores are 100/88.75/20/82.50%; normalized-index scores are 100/88.75/40/87.25%. Both tables were frozen before these inputs were generated, and the earlier selection rows are excluded. This confirmation compares the two placements; the Native/YaRN baseline panel uses a different, earlier input set.

Table 26: Independent-input placement comparisons, 80 examples per task and four tasks per panel. Intervals are for index minus direct-gap, in percentage points: 97.5% for each Qwen length (Bonferroni over two lengths), 95% for Gemma.

Model	Length	Direct-gap (%)	Index (%)	Difference interval
Qwen-0.5B	32K	49.69	53.37	$[-0.25, +7.75]$
Qwen-0.5B	64K	46.62	46.13	$[-4.87, +3.81]$
Gemma-2B	16K	72.81	79.00	$[+2.81, +9.62]$

F.3 BOUNDARY-MATCHED INTERMEDIATE EXPONENTS

Let l be the last native frequency pair making more than 32 turns over the reference window and h the first making less than one turn. Write $N = h - l$ and $q = \text{clip}(k - l, 0, N)$. All profiles leave the fast band unchanged and divide the slow-band frequencies by s :

$$\omega'_k = \omega_k^N s^{-m_q}, \quad m_0 = 0, \quad m_N = 1. \quad (80)$$

With the same band boundaries, the three intermediate allocations are

$$m_q^{\text{Uni}} = \frac{q}{N}, \quad (81)$$

$$m_q^{\text{Pro}} = \frac{q(q+1)}{N(N+1)}, \quad (82)$$

$$m_q^{\text{BM}} = \frac{q(q+1)(3N+2-2q)}{N(N+1)(N+2)}. \quad (83)$$

The Uni and Pro rules follow the uniform and progressive mixed-radix constructions of Tian et al. (2026). BM smooths the radix increments $\epsilon_q = m_q - m_{q-1}$ at both band boundaries.

To obtain BM, minimize $\sum_{q=0}^N (\epsilon_{q+1} - \epsilon_q)^2$ with $\epsilon_0 = \epsilon_{N+1} = 0$ and $\sum_{q=1}^N \epsilon_q = 1$. The discrete Euler equation gives

$$\epsilon_q = \frac{6q(N+1-q)}{N(N+1)(N+2)};$$

summing gives (83). The quadratic form is strictly convex with the endpoints fixed, so this solution is unique. Figure 9 shows the actual profile shapes for the tested transition widths.

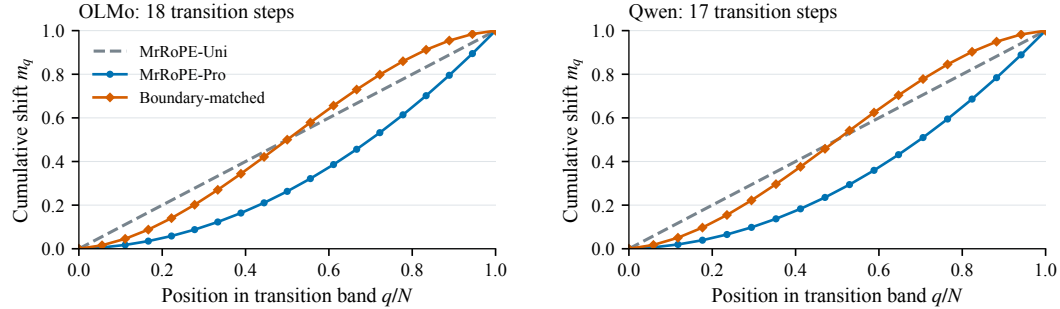


Figure 9: **Intermediate-band exponent displacements.** All three profiles have the same cumulative scale at the endpoints. BM redistributes the increments toward the middle of the transition band.

OLMo-2-0425-1B-Instruct uses $b = 500,000$, reference length 4096, $K = 64$, and $(l, h, N) = (14, 32, 18)$. Qwen-2.5-3B/7B-Instruct use $b = 10^6$, reference length 32768, $K = 64$, and $(l, h, N) = (23, 40, 17)$. The factor-four comparisons use $g = 1 + 0.1 \log 4 = 1.1386294361$ for both MrRoPE-Pro and BM, with FP32 frequency tables and BF16 model execution.

F.4 SIX-TASK MODEL COMPARISONS

The shared task types are NIAH single-2, multikey-2, multiquery, variable tracking, frequent words, and QA-1. OLMo’s confirmation uses seed 20260910 with four short and eight long examples per task, after a separate development panel with seed 20260909. Its 4K/16K scores are listed in Table 5; MrRoPE-Uni on the same confirmation inputs scores 76.88/32.12%, and official YaRN scores 54.38/6.94%. The Qwen-3B panel has two 32K and four 128K examples per task. The 7B screen reuses the first one short and two long examples per task; this selection was fixed before its outputs were inspected. Table 27 reports every task mean.

All arms use the corresponding native chat template and the same task-specific generation caps, score the entire saved response with the RULER task metric, and retain generated token IDs and EOS flags. On Qwen-7B, a tokenwise MLP chunk size of 4096 makes 128K evaluation fit in memory; both arms share that path while retaining full attention.

Table 27: All six-task BM/MrRoPE-Pro comparisons. Values are official RULER task scores (%); model-specific sample counts are given in Table 5.

Model	Task	Short		Long	
		MrPro	BM	MrPro	BM
OLMo-1B	Single-2	75.00	100.00	0.00	100.00
	MultiKey-2	25.00	100.00	0.00	37.50
	MultiQuery	18.75	87.50	0.00	43.75
	Variable tracking	25.00	70.00	0.00	10.00
	Frequent words	58.33	58.33	16.67	41.67
	QA-1	25.00	75.00	0.00	75.00
Qwen-3B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	75.00	50.00
	MultiQuery	100.00	100.00	93.75	100.00
	Variable tracking	90.00	100.00	75.00	75.00
	Frequent words	83.33	100.00	75.00	75.00
	QA-1	50.00	50.00	50.00	25.00
Qwen-7B	Single-2	100.00	100.00	100.00	100.00
	MultiKey-2	100.00	100.00	100.00	100.00
	MultiQuery	100.00	100.00	100.00	100.00
	Variable tracking	100.00	80.00	90.00	60.00
	Frequent words	100.00	100.00	66.67	66.67
	QA-1	0.00	0.00	50.00	0.00

F.5 NATURAL QUESTION ANSWERING

The natural comparison reuses frozen, source-verified LongBench prompts (Bai et al., 2024). Inputs are untruncated, retain their source questions and references, and fit the 16K input-plus-generation budget. Selection follows the pre-existing source-hash order; no generated answer participates in selection. The complete eligible pool contains 778 questions, of which 631 have more than 4096 input tokens. Both methods retain the same factor-four table at every length.

Greedy generation uses a 32-token cap for HotpotQA and 2WikiMQA, 128 for Qasper and NarrativeQA, and 64 for MultiFieldQA-en. The metric is token F1 on the complete saved generated response against the task references. Table 28 gives the long-input and original-window-length strata, including EOS counts. The long-input macro averages the five task means; the shorter-input macro averages its four available tasks. The respective BM-minus-MrPro changes are +3.8194 and +2.0028 points, with paired within-task bootstrap intervals [1.3161, 6.2920] and [-3.5618, 7.4742]. The MultiFieldQA long-input row includes 40 capped responses in each arm; these remain in the F1 calculation.

G SUPPORTING RESULTS

G.1 MULTI-HEAD LATENT ATTENTION

We report allocation at one fixed rotary budget in a 432M-parameter transformer following the MLA architectural pattern (DeepSeek-AI, 2024a), with 24 layers, width 1024, 16 heads, and per-head width 64. Its decoupled rotary subspace has $d_{\text{rope}}=32$ ($K=16$ frequencies); the remaining $d_{\text{nope}}=32$ dimensions are non-rotary, and the shared KV latent rank is 256. The model trains from scratch on 500M FineWeb-Edu tokens (Penedo et al., 2024) at $L_{\text{train}}=8192$ and base 500K over seeds 42, 43, and 88. EVQ-COSH uses the pre-specified $\tau = 1.414$ setting. At 16K, Geo PPL is 141.1/132.5/142.8 for seeds 42/43/88, whereas EVQ-COSH is 93.7/92.7/100.3; all three paired directions agree.

For the supporting overlay, let $\lambda_k = 2\pi/\omega_k$ and $t_k = \text{clip}((\lambda_k - L_{\text{train}})/(31L_{\text{train}}), 0, 1)$. The MLA wavelength-blend operator sets $\omega'_k = \omega_k/[1 + t_k(s - 1)]$: it preserves wavelengths below the training length, applies the full factor s above $32L_{\text{train}}$, and interpolates between them.

Table 28: **OLMo-2 Natural QA under static extension** ($s = 4$, $4K \rightarrow 16K$, $g = 1.1386$). Evaluation on allenai/OLMo-2-0425-1B-Instruct across 778 frozen LongBench inputs (greedy decoding, whole response token F1). **X2 Baseline Scope:** Boundary Matching (BM) is compared directly against MrRoPE-Pro on this exact panel, improving extended macro F1 by +3.82 pp (95% bootstrap CI [+1.32, +6.29]) and winning all five tasks; comparison against official YaRN is registered as a planned baseline.

Context Stratum	Task	n	MrPro F1 (%)	BM F1 (%)	Δ (pp)	MrPro EOS	BM EOS
Extended (4K–16K)	HotpotQA	166	28.63	33.54	+4.91	160	163
	2WikiMQA	173	21.55	25.43	+3.88	169	171
	Qasper	119	13.94	19.06	+5.11	119	119
	NarrativeQA	61	9.51	11.48	+1.96	61	61
	MultiFieldQA	112	34.49	37.72	+3.23	72	72
	Task-Equal Macro	631	21.62	25.44	+3.82	581	586
In-Window ($\leq 4K$)	HotpotQA	5	58.33	51.67	−6.67	5	5
	2WikiMQA	26	34.62	44.98	+10.37	26	26
	Qasper	78	26.16	28.14	+1.98	78	78
	MultiFieldQA	38	50.94	53.27	+2.34	35	33
	Task-Equal Macro	147	42.51	44.51	+2.00	144	142

Table 29: MLA validation (432M, $d_{\text{rope}}=32$, 3-seed mean $\pm$ std). Rows marked “MLA wavelength-blend” use this run’s wavelength-blend operator defined above.

Method	PPL@8K	PPL@16K	PPL@24K	PPL@32K
Geo	35.4 $\pm$ 0.9	138.8 $\pm$ 5.5	241.5 $\pm$ 2.6	323.7 $\pm$ 5.2
Geo + MLA wavelength-blend ($s = 4$)	35.5 $\pm$ 0.9	117.9 $\pm$ 6.5	204.9 $\pm$ 6.9	278.5 $\pm$ 6.3
EVQ-COSH	35.8 $\pm$ 0.8	95.6$\pm$4.1	204.9 $\pm$ 14.7	291.6 $\pm$ 20.5
EVQ-COSH + MLA wavelength-blend ($s = 4$)	35.8 $\pm$ 0.8	71.1$\pm$4.1	153.2$\pm$12.5	236.6$\pm$15.9

EVQ-COSH reduces 16K PPL by 31.1% ($138.8 \rightarrow 95.6$), while 8K PPL changes by +0.9%. All three training seeds favour EVQ-COSH at 16K. The EVQ-COSH plus MLA wavelength-blend row has the lowest mean PPL at 16K and beyond, showing that the allocation also composes with this wavelength-dependent extension.